USiS — Complete System Specification

Field	Value
Product	USiS — University Student Information System
Document type	Complete System Specification (SRS + Architecture + Features + LLD)
Date	November 2026
Classification	Proprietary

Document contents

1. System Requirements Specification (SRS)
2. Architecture Design
3. Features & Requirements
4. Low-Level Design (LLD) Workflows

Part 1 — System Requirements Specification (SRS)

University Student Information System (USiS)

Field	Value
Document ID	SRS-USIS-001
Product	USiS (University Student Information System)
Repository	`WMSU/wmsu-platform` (`apps/enrollment-web` , `apps/base-api` module `enrollment`)
Version	1.0
Date	November 2026
Classification	Proprietary

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines functional and non-functional requirements for **USiS**, the comprehensive student information system managing the complete academic lifecycle from admissions through graduation — including student registration, enrollment, advising, scheduling, academic records, grade management, assessment coordination, clearance processing, faculty evaluation, and student support services.

1.2 Scope

In scope

- Student self-service portal: profile management, enrollment workflow, schedule viewing, assessment/payment status, certificate of registration (COR), grades, academic history, support request submission
- Faculty portal: class rosters, grade encoding/submission, advising sessions, academic dashboards
- Registrar portal: applicant management, student admission, enrollment approval workflows, academic structure configuration (programs, curricula, sections), registration oversight, comprehensive reporting
- Office staff portals: Admissions (entrance exam coordination, applicant tracking), Office of Student Affairs (OSA: discipline, organizations, activities), Guidance (counseling sessions, assessments), Student Health Services (medical records, clearance), Scholarship Office (awards, disbursements, monitoring)

- Internal integration APIs for UFMS (assessment/payment handoff) and TORMS (grade projections)
- University Platform Access SSO integration (platform code `enrollment`)
- Queue management system for student services
- Faculty evaluation module with anonymous submissions
- Clearance certificate generation with public QR verification
- Academic calendar and enrollment window management
- Real-time enrollment conflict detection and waitlist management

Out of scope

- Public guest application forms and entrance exam delivery (hosted in Unified Access Portal `/apply*`)
- Financial cashiering, official receipt issuance, general ledger posting (**UFMS**)
- Transcript of Records (TOR) issuance and alumni registry management (**TORMS**)
- Library, document management, inventory operations (**ULMIS**, **UDMS**, **UIMS**)
- Learning management system and course content delivery (external LMS integration ready)

1.3 Definitions

Term	Definition
USiS	University Student Information System — enrollment and academic lifecycle platform
COR	Certificate of Registration — official enrollment document issued to students
Assessment	Tuition and fee calculation prepared for an enrollment period and synchronized with UFMS for payment processing
Role pack	Collection of USiS permission codes mapped from University Platform Access Identity role
Enrollment workflow	State-based process from draft enrollment through advising, payment verification, registrar approval, to registered status
Clearance graph	Network of office-specific clearance requirements and status checks across university departments
Grade projection	Academic grade record synchronized from USiS to TORMS for transcript generation
Section	Class offering with instructor, schedule, room, and enrollment capacity

1.4 References

- `WMSU/wmsu-platform/README.md`
- `apps/web-portal/public/docs/USIS-Complete-Guide.html`
- `apps/web-portal/public/docs/USIS-API-Specs-Manual.html`
- University Platform Access Architecture (Identity integration)
- UFMS Internal API Specification (assessment handoff)
- TORMS Sync Protocol (grade projections)

2. Overall Description

2.1 Product Perspective

USiS operates within the WMSU Campus Information System (CIS) modular architecture:

Component	Technology	Role
Enrollment web	React 19 + Vite	Multi-role portal UI (student, faculty, registrar, offices)
Base API enrollment module	AdonisJS 7 domain module	REST services at `/api/v1/enrollment/*`
Database	PostgreSQL schema `enrollment`	Academic system of record (SoR)
Identity provider	University Platform Access	Authentication, authorization, MFA, role management
Queue system	BullMQ + Redis	Background jobs (notifications, batch processing, reports)

Surface	Dev port	Production access
Enrollment web	`http://127.0.0.1:5180`	`https://enrollment.wmsu.edu.ph` (example)
Base API	`http://127.0.0.1:3500`	`https://api.wmsu.edu.ph`
Identity API	`http://127.0.0.1:3411`	`https://auth.wmsu.edu.ph`
Portal launcher	`http://127.0.0.1:5173`	`https://portal.wmsu.edu.ph`

Health endpoint: `GET /health` → `{ status: 'ok', service: 'base-api', module: 'enrollment' }`

2.2 User Classes

Role	Primary capabilities	Access level
Student	View profile, enroll in courses, view schedules/grades, submit support requests, download COR, track payments, evaluate faculty	Own data only
Faculty	View assigned classes, encode/submit grades, conduct advising sessions, generate class reports, manage attendance	Assigned classes
Adviser	Review advisee enrollments, approve/recommend course selections, monitor academic progress	Assigned students
Registrar	Manage applicants, approve enrollments, configure academic structures, oversee all registrations, generate institutional reports	System-wide academic data

Admissions Officer	Manage entrance exam schedules, process applications, track admission workflows	Applicant records
OSA Staff	Manage student discipline, organization accreditations, event coordination	Student conduct records
Guidance Counselor	Schedule counseling sessions, maintain confidential records, issue clearances	Counseling records
Health Services Staff	Maintain medical records, manage health clearances, track vaccinations	Student health records
Scholarship Officer	Manage scholarship awards, track disbursements, monitor academic requirements	Scholarship records
Cashier (read-only)	View assessment and payment status for student services	Payment status queries
Identity Administrator	Full system access when `enrollment` platform granted	All USiS permissions

Frontend permission checks provide user experience only; API enforces authorization on every authenticated route.

2.3 Constraints

- Platform tile visibility requires Identity platform grant `enrollment`
- Menu and feature access controlled by role-specific permission packs
- Guest application forms redirect to Portal; USiS does not host public `/apply*` routes
- Payment verification strictly depends on UFMS internal APIs — USiS never invents "paid" status
- Grade system of record is USiS; TORMS consumes projections read-only
- Multi-campus university structure with hierarchical org units (University → Campus → College → Department → Program)
- Academic calendar windows control enrollment availability
- Real-time section capacity enforcement prevents over-enrollment
- Prerequisite and co-requisite validation enforced at enrollment time
- Assessment amount calculation follows configured fee schedules — amounts not hard-coded in SRS

2.4 Assumptions

- University Platform Access Identity and Base API are running and reachable
- Demo/staging environments provision seed users, academic structures, and sample enrollments
- Fee schedules and assessment rules maintained in UFMS configuration — not independently managed in USiS
- Entrance exam scores and applicant data synced from Portal admission module
- Academic year and term structures configured before enrollment windows open
- Room and schedule data available for conflict checking

- Email/SMS notification infrastructure configured for enrollment alerts

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3. Functional Requirements

3.1 Authentication & Authorization

ID	Requirement	Priority
FR-AUTH-001	System shall authenticate all users via University Platform Access SSO JWKS verification	Must
FR-AUTH-002	System shall resolve user home path based on role packs (student → `/student`, registrar → `/registrar`, faculty → `/faculty`)	Must
FR-AUTH-003	System shall enforce permission checks on every API endpoint using role-specific permission codes	Must
FR-AUTH-004	System shall support MFA for privileged roles (registrar, office heads) when Identity enforces	Must
FR-AUTH-005	System shall provide graceful error handling when Identity service unavailable	Must

3.2 Student Portal

ID	Requirement	Priority
FR-STU-001	Students shall view and update demographic profile information with field-level validation	Must
FR-STU-002	Students shall initiate enrollment for available academic terms within enrollment windows	Must
FR-STU-003	Students shall search available sections by subject, instructor, schedule, and available seats	Must
FR-STU-004	Students shall add sections to enrollment with real-time conflict detection (schedule, prerequisite)	Must
FR-STU-005	Students shall view total units and enforcement of minimum/maximum load limits	Must
FR-STU-006	Students shall submit enrollment for advising approval when required by program	Must
FR-STU-007	Students shall view assessment amounts after enrollment submission with line-item breakdown	Must
FR-STU-008	Students shall view payment status synchronized from UFMS without locally caching paid state	Must

FR-STU-009	Students shall download PDF Certificate of Registration (COR) after enrollment approval	Must
FR-STU-010	Students shall view class schedules in calendar and list formats with room/instructor details	Must
FR-STU-011	Students shall view posted grades by term with GPA calculation	Must
FR-STU-012	Students shall view academic history including all prior enrollments and grades	Must
FR-STU-013	Students shall submit support requests to designated offices (OSA, Guidance, Health, Scholarship)	Must
FR-STU-014	Students shall submit organization accreditation applications with file attachments	Should
FR-STU-015	Students shall request student ID cards through portal	Should
FR-STU-016	Students shall obtain queue tickets for office services	Should
FR-STU-017	Students shall evaluate faculty anonymously during evaluation windows	Must
FR-STU-018	Students shall view clearance status with office-by-office breakdown	Must
FR-STU-019	Students shall download clearance certificates with QR verification codes	Must

3.3 Faculty Portal

ID	Requirement	Priority
FR-FAC-001	Faculty shall view assigned class sections with enrollment rosters	Must
FR-FAC-002	Faculty shall encode preliminary and final grades for enrolled students	Must
FR-FAC-003	Faculty shall validate grade completeness before submission	Must
FR-FAC-004	Faculty shall submit grades with digital signature/confirmation	Must
FR-FAC-005	Faculty shall view grade submission history and status	Must
FR-FAC-006	Faculty shall request grade corrections with justification after submission	Should
FR-FAC-007	Faculty shall download class rosters as CSV and PDF	Must
FR-FAC-008	Faculty shall mark student attendance by session	Should
FR-FAC-009	Faculty shall view advisee list with enrollment status	Must

FR-FAC-010	Faculty shall approve/reject advisee enrollment plans	Must
FR-FAC-011	Faculty shall add advising notes and recommendations	Must
FR-FAC-012	Faculty shall view evaluation summary results (anonymized)	Should

3.4 Registrar Portal

ID	Requirement	Priority
FR-REG-001	Registrar shall view and search applicant records from admission pipeline	Must
FR-REG-002	Registrar shall admit applicants with student number assignment and Identity account provisioning	Must
FR-REG-003	Registrar shall view all student enrollments with filtering by term, program, status	Must
FR-REG-004	Registrar shall approve or reject enrollment submissions with reason codes	Must
FR-REG-005	Registrar shall perform batch enrollment approval with selection criteria	Should
FR-REG-006	Registrar shall override enrollment business rules with audit logging	Must
FR-REG-007	Registrar shall manage academic calendar with term dates and enrollment windows	Must
FR-REG-008	Registrar shall configure programs, curricula, and courses	Must
FR-REG-009	Registrar shall define section offerings with instructor, schedule, room, capacity	Must
FR-REG-010	Registrar shall configure prerequisite and co-requisite rules	Must
FR-REG-011	Registrar shall configure grading systems and policies by program	Must
FR-REG-012	Registrar shall generate enrollment reports (summary by program, status, term)	Must
FR-REG-013	Registrar shall generate grade reports (distribution, dean's list, deficiency)	Must
FR-REG-014	Registrar shall export reports as CSV, Excel, and PDF	Must
FR-REG-015	Registrar shall monitor grade submission status across all sections	Must

FR-REG-016	Registrar shall initiate faculty evaluation periods by term	Must
FR-REG-017	Registrar shall view aggregated evaluation results with filtering	Should
FR-REG-018	Registrar shall manage waitlists and process seat releases	Should

3.5 Office Portals

ID	Requirement	Priority
FR-OFF-001	Admissions officers shall schedule entrance exam sessions with capacity management	Must
FR-OFF-002	Admissions officers shall view exam results and track admission decisions	Must
FR-OFF-003	OSA staff shall manage student discipline cases with confidential records	Must
FR-OFF-004	OSA staff shall process organization accreditation applications with approval workflow	Must
FR-OFF-005	OSA staff shall coordinate campus events and student participation tracking	Should
FR-OFF-006	Guidance counselors shall schedule confidential counseling sessions	Must
FR-OFF-007	Guidance counselors shall maintain counseling records with restricted access	Must
FR-OFF-008	Guidance counselors shall issue guidance clearances	Must
FR-OFF-009	Health services staff shall maintain student medical records	Must
FR-OFF-010	Health services staff shall track vaccinations and health requirements	Must
FR-OFF-011	Health services staff shall issue health clearances	Must
FR-OFF-012	Scholarship officers shall manage scholarship awards by type and amount	Must
FR-OFF-013	Scholarship officers shall track disbursement schedules	Must
FR-OFF-014	Scholarship officers shall monitor scholarship academic requirements	Must
FR-OFF-015	All offices shall contribute to student clearance status graph	Must

3.6 Assessment & Payment Integration

ID	Requirement	Priority
FR-PAY-001	System shall prepare tuition assessment documents from enrollment data	Must
FR-PAY-002	System shall synchronize assessment data to UFMS via internal API	Must
FR-PAY-003	System shall query UFMS for payment verification status	Must
FR-PAY-004	System shall block enrollment progression when payment pending per business rules	Must

FR-PAY-005	System shall never locally cache or invent "payment verified" status	Must
FR-PAY-006	System shall display assessment line items from configured fee schedules	Must
FR-PAY-007	System shall support installment payment plans tracked in UFMS	Should

3.7 Grade Management & TORMS Sync

ID	Requirement	Priority
FR-GRD-001	System shall enforce grade encoding deadlines by academic calendar	Must
FR-GRD-002	System shall validate grade values against configured grading system	Must
FR-GRD-003	System shall calculate GPA using weighted credit hours	Must
FR-GRD-004	System shall lock submitted grades from faculty modification	Must
FR-GRD-005	System shall process registrar-approved grade corrections with audit trail	Must
FR-GRD-006	System shall expose grade projection API for TORMS consumption	Must
FR-GRD-007	System shall maintain grade history for all correction cycles	Must

3.8 Clearance Management

ID	Requirement	Priority
FR-CLR-001	System shall maintain configurable clearance requirements by office and term	Must
FR-CLR-002	System shall aggregate clearance status from multiple office systems	Must
FR-CLR-003	System shall generate clearance certificates with QR verification tokens	Must
FR-CLR-004	System shall provide public unauthenticated clearance verification endpoint	Must
FR-CLR-005	System shall enforce clearance requirements for enrollment, graduation, and document issuance	Must

3.9 Faculty Evaluation

ID	Requirement	Priority
FR-EVL-001	System shall configure evaluation periods by term with questionnaire templates	Must
FR-EVL-002	System shall require students to evaluate enrolled faculty within evaluation window	Must
FR-EVL-003	System shall enforce anonymous submission (no linkage to student identity in results)	Must

FR-EVL-004	System shall aggregate evaluation responses with statistical summaries	Must
FR-EVL-005	System shall restrict raw evaluation access to authorized administrators	Must
FR-EVL-006	Faculty shall view anonymized aggregated results for own sections	Should

3.10 Reporting & Analytics

ID	Requirement	Priority
FR-RPT-001	System shall generate enrollment summary reports by term, program, level	Must
FR-RPT-002	System shall generate grade distribution reports by course and section	Must
FR-RPT-003	System shall generate dean's list and academic probation lists	Must
FR-RPT-004	System shall generate graduation candidate lists	Must
FR-RPT-005	System shall export all reports in CSV, Excel, and PDF formats	Must
FR-RPT-006	System shall log all report exports with user and timestamp for audit	Must

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4. External Interface Requirements

4.1 Integration Interfaces

Interface	Direction	Protocol	Purpose
University Platform Access	Inbound	OIDC/JWKS	SSO authentication, JWT verification, role resolution
UFMS Finance APIs	Outbound	REST+JSON	Assessment upsert, payment status query, clearance verification
TORMS Records APIs	Outbound	REST+JSON	Grade projection sync, academic history provision
Portal Admissions	Inbound	REST+JSON	Applicant data sync from entrance exam and application system
SMTP/SMS Gateway	Outbound	SMTP/HTTP	Enrollment notifications, grade alerts, deadline reminders
Object Storage	Outbound	S3-compatible	Document attachments (support requests, ID photos)

4.2 API Specifications

Enrollment API base: `/api/v1/enrollment/*`

OpenAPI documentation: Available at `/api/v1/enrollment/docs` (Swagger UI)

Authentication: Bearer JWT from University Platform Access in `Authorization` header

Internal service routes: `/api/v1/enrollment/internal/*` protected by `X-Service-Api-Key` header

4.3 User Interface Requirements

- Responsive design supporting desktop (≥1280px), tablet (768-1023px), and mobile (360-767px)
- Browser compatibility: Chrome/Edge (latest 2 versions), Firefox (latest 2), Safari (latest 2)
- Accessibility: WCAG 2.1 Level AA conformance for student-facing flows
- Print-optimized views for COR, schedules, clearance certificates
- Real-time updates for enrollment conflicts and seat availability
- Progressive enhancement for offline-capable features (view cached COR/schedule)

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5. Non-Functional Requirements

5.1 Performance

ID	Requirement	Target
NFR-PERF-001	API response time (p95) for standard queries	≤ 2 seconds
NFR-PERF-002	API response time (p95) for enrollment submission	≤ 5 seconds
NFR-PERF-003	Concurrent user support during enrollment peak	≥ 500 active sessions
NFR-PERF-004	Health check endpoint response time	≤ 500 ms
NFR-PERF-005	Report generation for standard queries	≤ 10 seconds
NFR-PERF-006	Real-time conflict detection during enrollment	≤ 1 second

5.2 Scalability

ID	Requirement
NFR-SCAL-001	Support ≥15,000 enrolled students per term
NFR-SCAL-002	Support ≥1,000 concurrent enrollment transactions
NFR-SCAL-003	Horizontal scaling of API replicas without session affinity
NFR-SCAL-004	Database read replicas for reporting queries

5.3 Security

ID	Requirement

NFR-SEC-001	TLS 1.2+ required for all client-server communication
NFR-SEC-002	RBAC enforcement on every authenticated API endpoint
NFR-SEC-003	Service-to-service communication via API keys for internal routes
NFR-SEC-004	Sensitive data (grades, discipline, counseling) encrypted at rest
NFR-SEC-005	Password policy and MFA enforcement delegated to Identity service
NFR-SEC-006	CORS allowlist limited to authorized web client origins
NFR-SEC-007	Rate limiting on public clearance verification endpoints
NFR-SEC-008	Audit logging of privileged operations (enrollment override, grade corrections)
NFR-SEC-009	Non-root container execution in production
NFR-SEC-010	Regular dependency vulnerability scanning and patching

5.4 Reliability

ID	Requirement
NFR-REL-001	System availability ≥99.5% during enrollment periods
NFR-REL-002	Automated database backups (daily incremental, weekly full)
NFR-REL-003	Point-in-time recovery capability ≤15 minutes RPO
NFR-REL-004	Documented disaster recovery procedures with quarterly drill
NFR-REL-005	Graceful degradation when UFMS/TORMS temporarily unavailable

5.5 Maintainability

ID	Requirement
NFR-MAIN-001	Modular domain architecture with clear separation of concerns
NFR-MAIN-002	Comprehensive API documentation (OpenAPI 3.0+)
NFR-MAIN-003	Structured logging with correlation IDs for request tracing
NFR-MAIN-004	Observability via Sentry/OpenTelemetry when configured
NFR-MAIN-005	Database schema migrations versioned and reversible

5.6 Compliance

ID	Requirement
NFR-COMP-001	Data privacy compliance with RA 10173 (Data Privacy Act of 2012)

NFR-COMP-002	Purpose limitation for student personal data collection
NFR-COMP-003	Student consent management for optional data uses
NFR-COMP-004	Data retention policies aligned with university records management
NFR-COMP-005	FERPA-equivalent protections for educational records

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6. Verification Criteria

6.1 Functional Verification

Scenario	Expected result	Test type
SSO login as student	Redirect to `/student` with profile data	Integration
SSO login as registrar	Redirect to `/registrar` with system dashboard	Integration
Enroll in section with prerequisite not met	Error message with missing prerequisite details	Unit
Enroll in section with schedule conflict	Error message with conflicting section details	Unit
Submit enrollment for advising	Status changes to "pending_advising"	Integration
Faculty encode grades	Grades saved; submission blocked until complete	E2E
Faculty submit grades	Grades locked; visible to student	E2E
Query UFMS for verified payment	Enrollment progresses to registrar approval	Integration
Query UFMS for pending payment	Enrollment blocked with payment required message	Integration
Generate clearance certificate	PDF with QR code generated	E2E
Verify valid clearance QR	Public endpoint returns valid clearance data	API test
Verify invalid clearance QR	Public endpoint returns not found / invalid	API test
Sync grades to TORMS	TORMS receives grade projections via internal API	Integration

6.2 Non-Functional Verification

Scenario	Expected result	Test type
500 concurrent enrollments	p95 response time ≤5s, no errors	Load test
Unauthorized API access	HTTP 401/403 with error message	Security
Horizontal scaling test	Add replica; traffic balanced; session independent	Scalability
Database failover	Recovery within 15 minutes with data integrity	DR drill

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7. Document Control

Version	Date	Description
1.0	November 2026	Complete System Specification from wmsu-platform enrollment module

Related Design Documents

- **USiS-Architecture.md** — Architecture design (containers, domains, security, deployment)
- **USiS-Features-Requirements.md** — Feature catalog + requirements traceability matrix
- **USiS-LLD.md** — Low-level design workflows (enrollment, grading, advising, clearance, UFMS/TORMS integration)

End of SRS — USiS

Part 2 — Architecture Design

USiS — University Student Information System Architecture

Field	Value
Document ID	AD-USIS-001
Product	USiS
Companions	USiS-SRS.md · USiS-Features-Requirements.md · USiS-LLD.md
Date	November 2026

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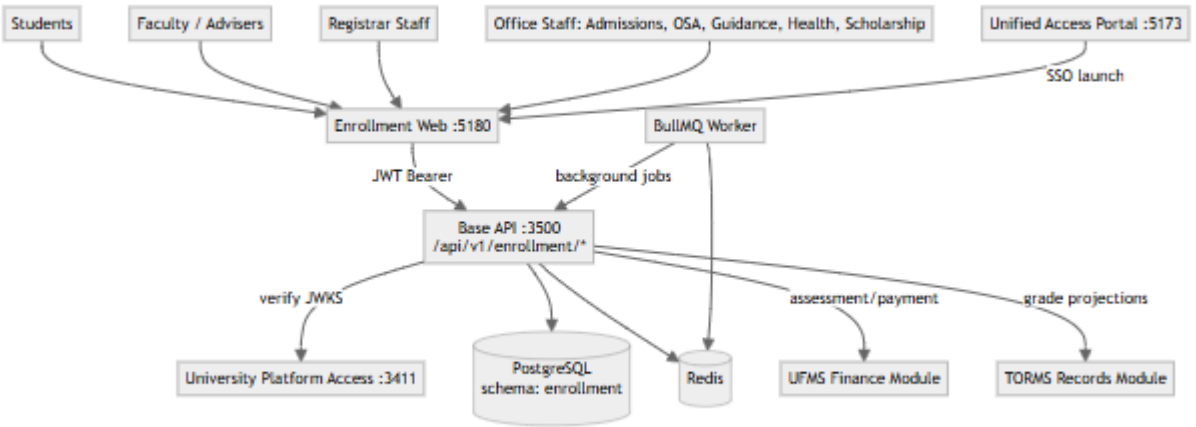
1. Purpose

Architecture design for USiS within the WMSU Campus Information System modular monolith — covering system context, container architecture, domain module organization, security architecture, deployment topology, integration patterns, and quality attributes.

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2. System Context

USiS operates within the CIS ecosystem, integrating with Identity, UFMS, TORMS, and the Portal:



Key relationships:

- Students, faculty, and staff authenticate through Portal SSO, launching enrollment web
- All UI interactions flow through Base API enrollment module endpoints

- Identity service provides JWT verification and role resolution
- UFMS owns financial transactions; USiS queries payment status
- TORMS consumes grade projections; USiS remains academic SoR
- Background jobs (notifications, reports) processed by BullMQ worker

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3. Container View

Container	Technology	Port	Purpose
Enrollment web	React 19 + Vite	5180	Multi-role portal UI (student, faculty, registrar, offices) with responsive design
Base API (enrollment module)	AdonisJS 7 + Lucid ORM	3500	Domain services, controllers, validators, business logic
PostgreSQL	PostgreSQL 15+	5432	Schema `enrollment` — academic system of record
Redis	Redis 7+	6379	Session cache, queue backend, real-time data
BullMQ Worker	Node.js + BullMQ	N/A	Background job processor (notifications, reports, sync)
University Platform Access	Identity service	3411	Authentication, authorization, MFA, user management
Unified Access Portal	React SPA	5173	SSO launcher and guest admissions UI

Health monitoring: `GET /health` returns service status and active module checks.

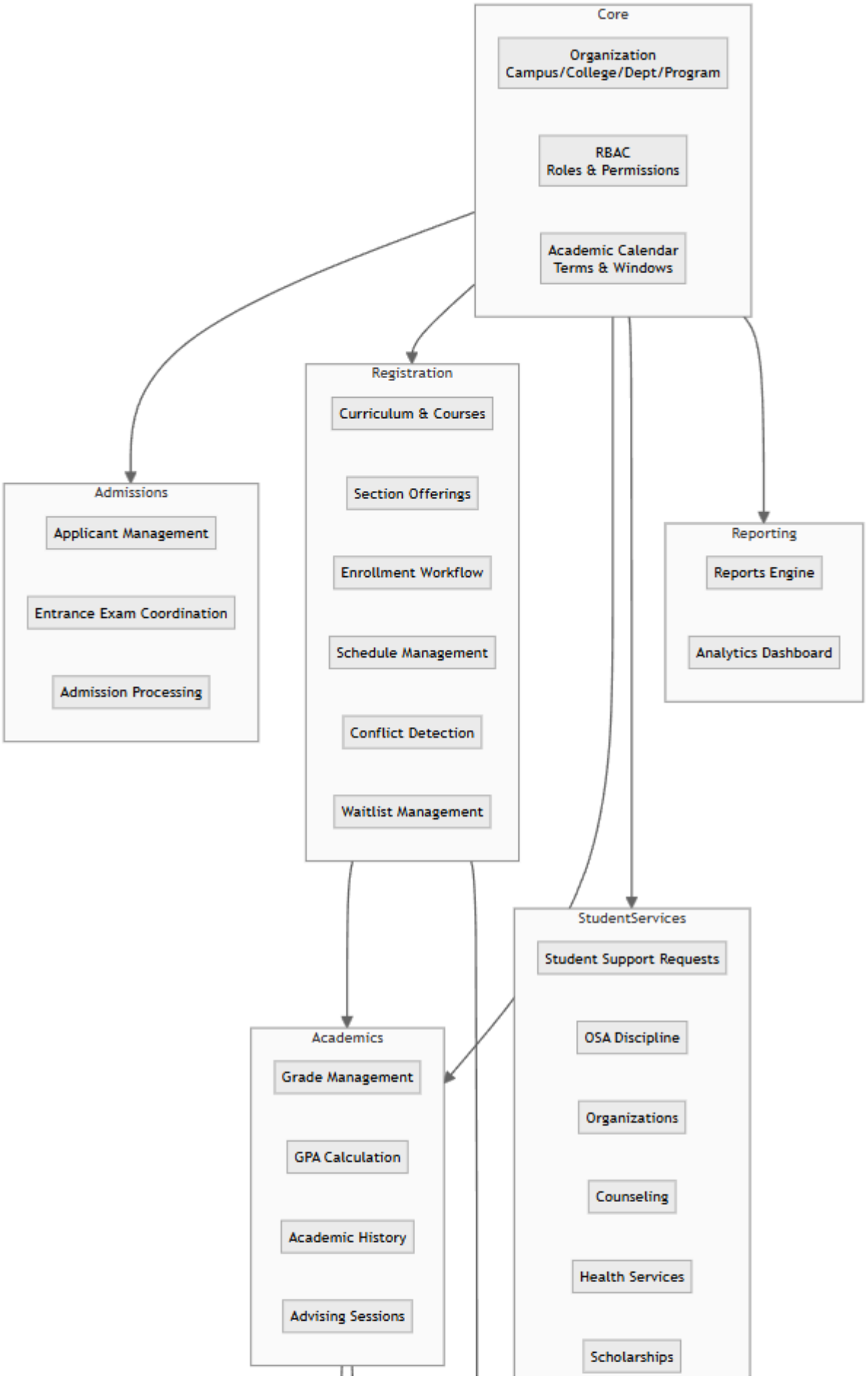
Deployment notes:

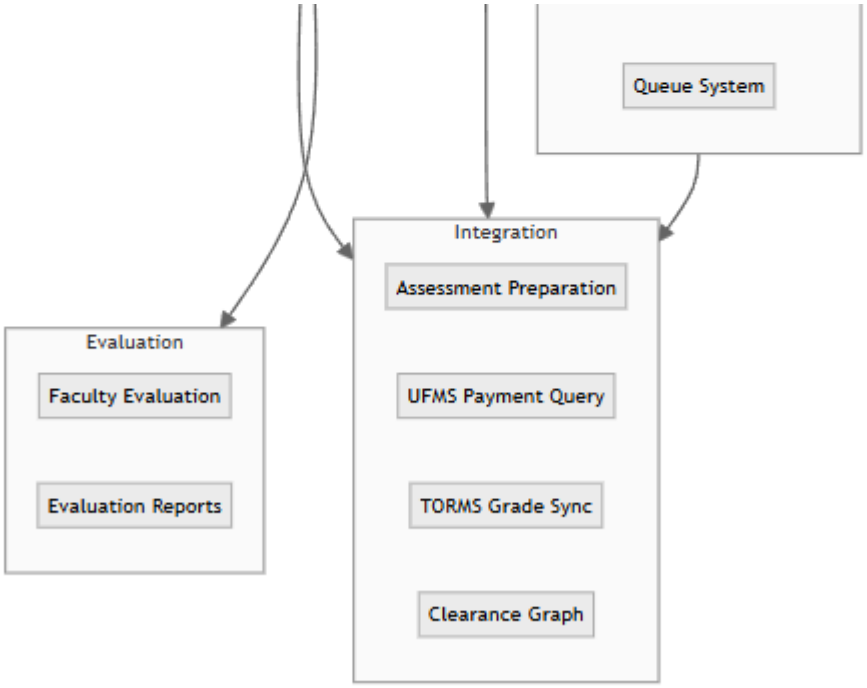
- Enrollment web statically served from CDN or Vite dev server
- Base API runs in containerized Node.js (non-root)
- Postgres uses schema isolation (`enrollment`) within shared `wmsu_cis` database
- Redis configured for persistence (AOF + RDB) for queue durability
- Worker deployed as separate container/process pool

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4. Domain Module Architecture

USiS enrollment module organized by academic lifecycle domains:


Core
Organization
Campus/College/Dept/Program
RBAC
Roles & Permissions
Academic Calendar
Terms & Windows
Admissions
Applicant Management
Entrance Exam Coordination
Admission Processing
Registration
Curriculum & Courses
Section Offerings
Enrollment Workflow
Schedule Management
Conflict Detection
Waitlist Management
Reporting
Reports Engine
Analytics Dashboard
StudentServices
Student Support Requests
OSA Discipline
Organizations
Counseling
Health Services
Scholarships
Academics
Grade Management
GPA Calculation
Academic History
Advising Sessions



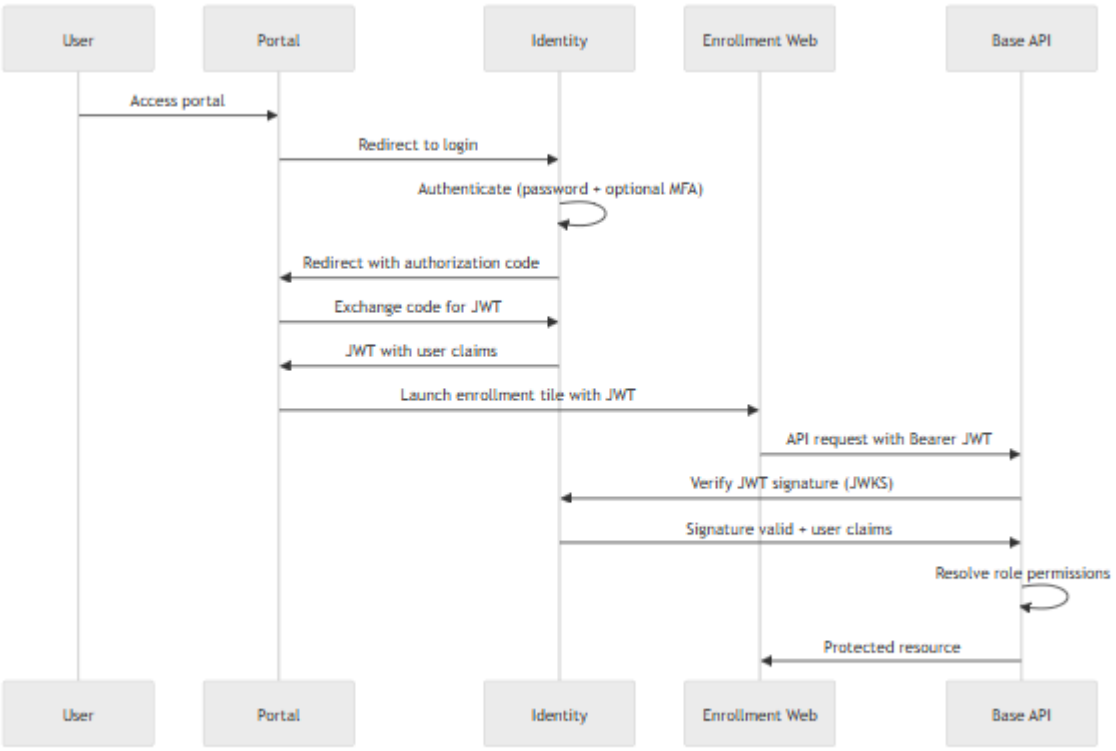
Domain responsibilities:

Domain	Key entities	Services
Core	Organization, Campus, College, Department, Program, AcademicTerm, EnrollmentWindow	Org hierarchy, calendar management, RBAC
Admissions	Applicant, EntranceExam, AdmissionDecision	Applicant tracking, exam scheduling, admission workflow
Registration	Curriculum, Course, Section, Enrollment, EnrollmentItem, Prerequisite	Section offerings, enrollment workflow, conflict detection, waitlist
Academics	Grade, GradeSubmission, AcademicRecord, AdvisingSession	Grade encoding, submission, GPA calculation, advising
Student Services	SupportRequest, DisciplineCase, Organization, CounselingSession, HealthRecord, Scholarship	Multi-office student support, clearance contribution
Integration	Assessment, PaymentStatus, GradeProjection, ClearanceItem	UFMS/TORMS handoff, clearance aggregation
Evaluation	EvaluationPeriod, EvaluationResponse, EvaluationSummary	Anonymous faculty evaluation
Reporting	Report, ReportExport, Dashboard	Enrollment/grade reports, analytics

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5. Security Architecture

5.1 Authentication Flow



Security controls:

Layer	Control	Implementation
Transport	TLS 1.2+	Reverse proxy/ingress termination
Authentication	OIDC/JWT	Identity service RS256 tokens
Authorization	RBAC	Permission checks on every API endpoint
API protection	Rate limiting	Redis-backed rate limiter
Secrets	Environment variables	Never in codebase or client
Database	Schema isolation	`enrollment` schema, role-based PG access
Internal APIs	Service keys	`X-Service-Api-Key` header validation
Public endpoints	Rate limiting	Clearance verification endpoint

5.2 Permission Model

USiS defines granular permissions mapped to Identity roles:

Permission categories:

- `enrollment.*` — Enrollment CRUD and approval
- `grades.*` — Grade encoding, submission, corrections
- `students.*` — Student profile and records
- `sections.*` — Section management
- `advising.*` — Advising workflows
- `offices.*` — Office-specific permissions (OSA, Guidance, Health, Scholarship)
- `reports.*` — Report generation and export
- `evaluation.*` — Faculty evaluation management

Role examples:

- `student` role → `enrollment.own`, `grades.view_own`, `clearance.view_own`
- `faculty` role → `grades.encode`, `grades.submit`, `advising.manage`
- `registrar` role → `enrollment.approve`, `students.manage`, `reports.export`
- `osa_staff` role → `offices.osa.*`, `students.discipline`

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6. Data Architecture

6.1 Schema Organization

PostgreSQL schema **`enrollment`** contains all USiS tables:

Core tables:

- `organizations`, `campuses`, `colleges`, `departments`, `programs`
- `academic_terms`, `enrollment_windows`
- `users` (Identity reference cache)

Academic structure:

- `curricula`, `courses`, `prerequisites`, `sections`
- `instructors`, `room_schedules`

Enrollment lifecycle:

- `enrollments`, `enrollment_items`, `enrollment_history`
- `waitlists`

Grades:

- `grades`, `grade_submissions`, `grade_history`
- `academic_records` (computed)

Student services:

- `support_requests`, `discipline_cases`, `organizations`
- `counseling_sessions`, `health_records`, `scholarships`

Integration:

- `assessments`, `payment_status_cache` (UFMS)
- `grade_projections` (TORMS)
- `clearance_items`, `clearance_certificates`

Evaluation:

- `evaluation_periods`, `evaluation_responses`

Reporting:

- `report_exports` (audit log)

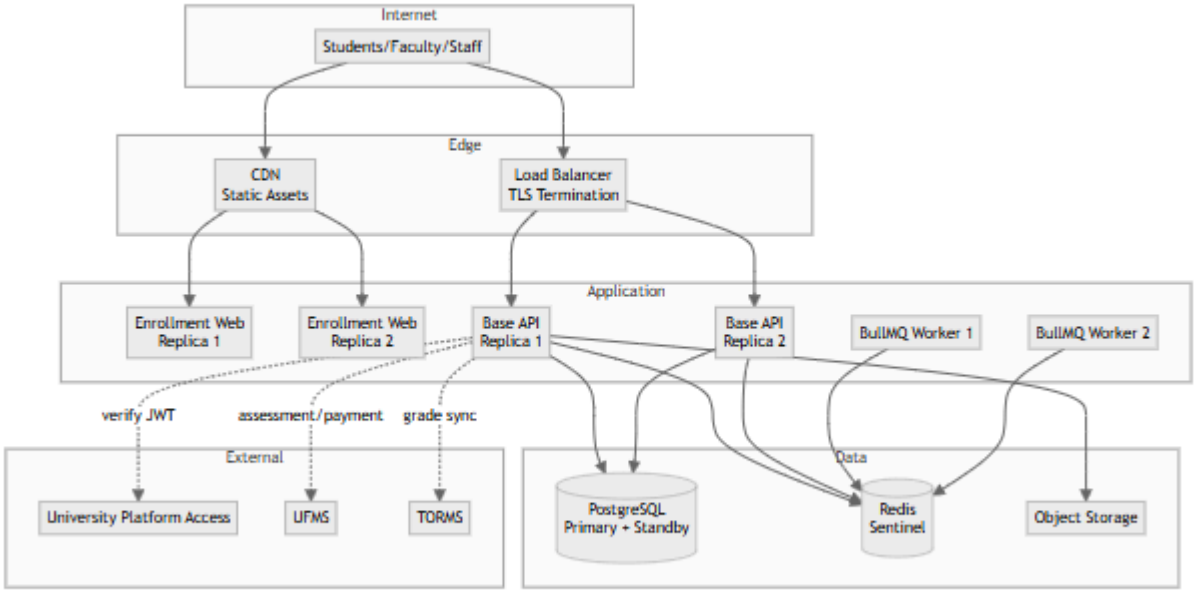
6.2 Migration Strategy

- Migrations tracked in `enrollment.uims_adonis_schema` table
- Schema changes deployed via `node ace migration:run`
- Rollback capability for reversible migrations
- Data seeds for demo/staging environments

---

7. Deployment Topology

7.1 Production Architecture



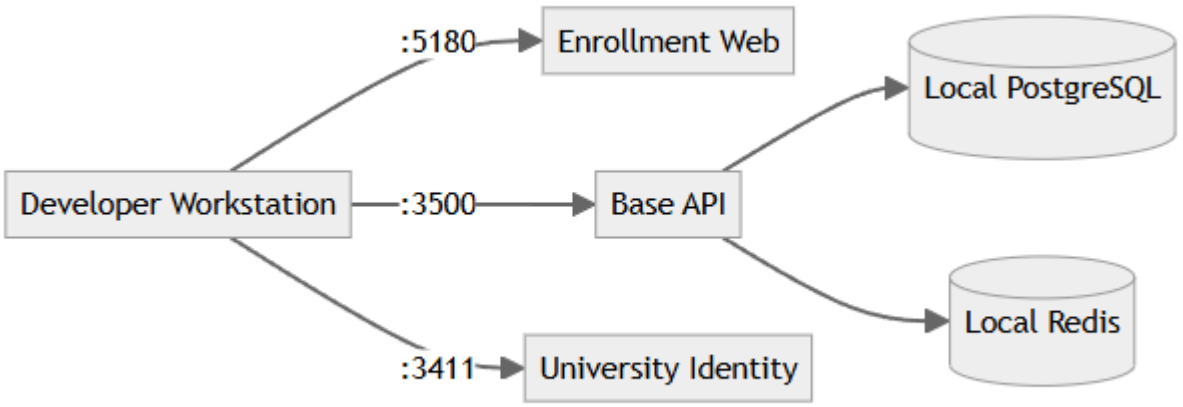
Sizing guidance (500 concurrent users):

- API: 2–3 replicas, 2 vCPU / 4 GB RAM each
- Worker: 2 replicas, 1 vCPU / 2 GB RAM each
- PostgreSQL: 4 vCPU / 16 GB RAM, managed service preferred
- Redis: 2 vCPU / 8 GB RAM, Sentinel for HA

Scalability:

- Stateless API enables horizontal scaling
- Read replicas for reporting queries
- CDN for static assets reduces origin load

7.2 Local Development



Local ports:

- Portal: 5173
- Identity: 3411
- Base API: 3500
- Enrollment web: 5180
- Finance web: 5182 (UFMS)
- Records web: 5181 (TORMS)

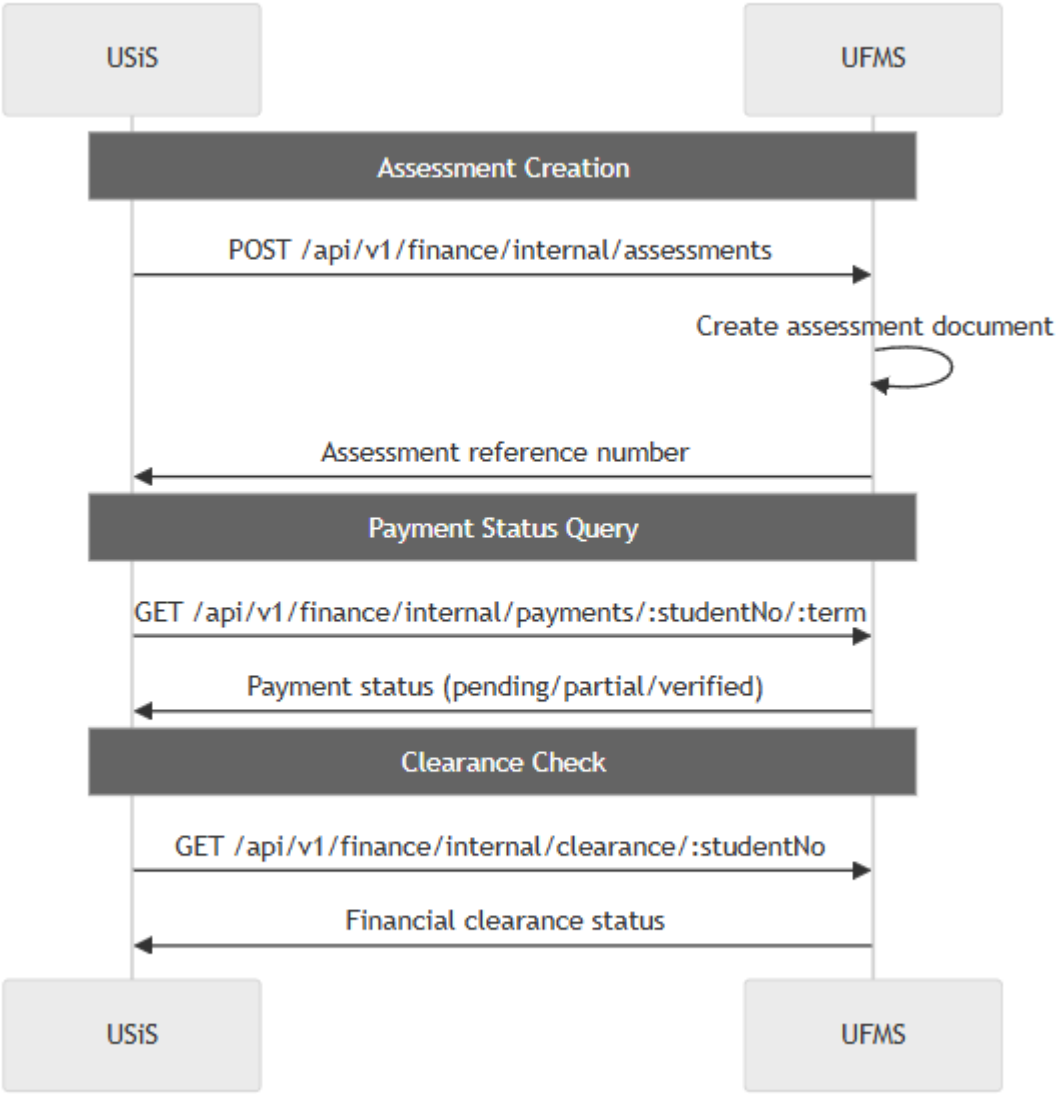
Infrastructure: Docker Compose for PostgreSQL, Redis, MinIO

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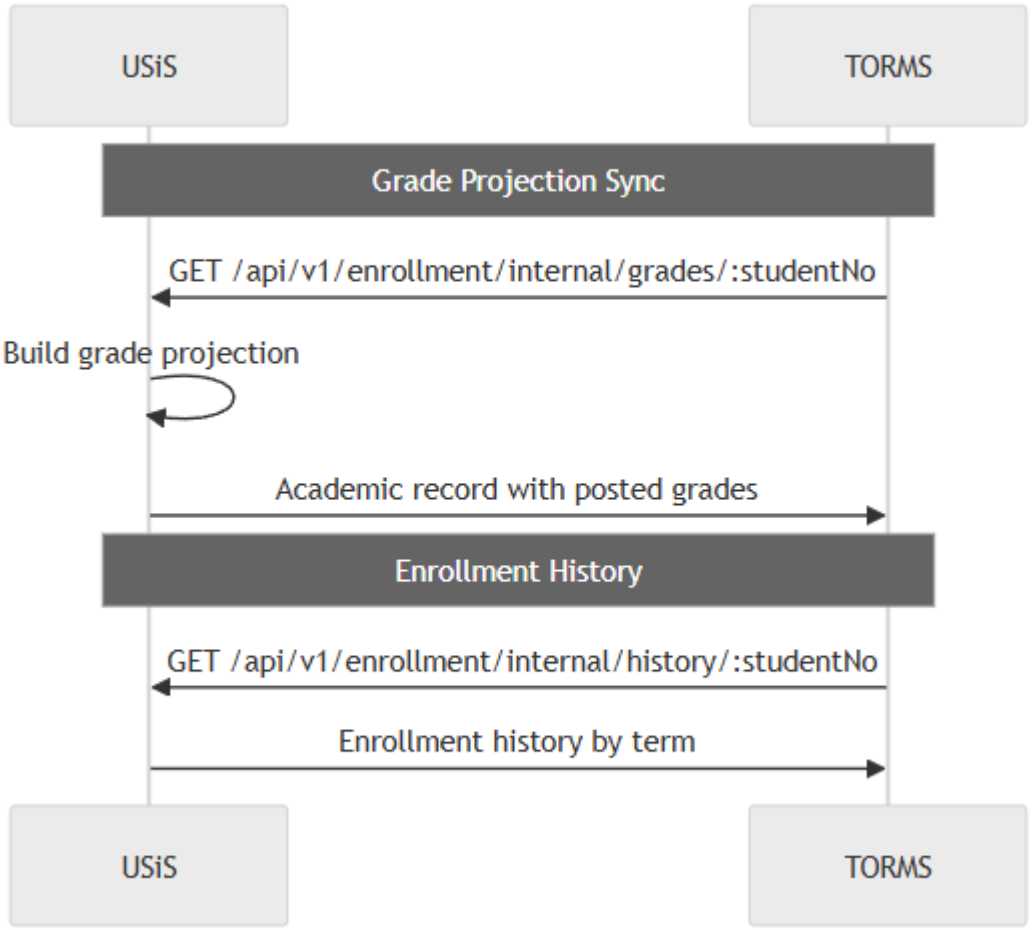
8. Integration Architecture

8.1 Internal Service Communication

UFMS Integration (Financial):



TORMS Integration (Academic Records):



Service-to-service security:

- Internal routes protected by `X-Service-Api-Key` header
- Service keys hashed in database, plaintext only in secure config
- Browser clients never receive service keys

8.2 External System Integration

System	Protocol	Pattern	Frequency
Portal Admissions	REST API	Applicant sync	Real-time webhook + nightly batch
SMTP Gateway	SMTP	Email notifications	Event-driven
SMS Gateway	HTTP API	SMS alerts	Event-driven
Object Storage (S3)	S3 API	File uploads	On-demand

---

9. Quality Attributes

9.1 Performance

Optimization strategies:

- Database indexing on frequently queried fields (student_no, term, status)
- Query optimization with Lucid eager loading to prevent N+1
- Redis caching for static reference data (programs, courses)
- Lazy loading of enrollment web routes
- Paginated API responses (default 25, max 100 items)
- Background jobs for heavy operations (report generation, bulk notifications)

Monitoring:

- APM (Sentry/OpenTelemetry) for request tracing
- Database slow query logging
- API endpoint metrics (response time, error rate)

9.2 Scalability

Horizontal scaling:

- Stateless API design (session state in Redis/JWT)
- Load balancer with health check probes
- Database connection pooling (PgBouncer)
- Queue-based job processing prevents request timeout

Vertical limits:

- Single PostgreSQL instance supports 15,000 students (tested)
- Read replicas for reporting offload main database

9.3 Reliability

Failure handling:

- Graceful degradation when UFMS/TORMS unavailable (cached status, error messaging)
- Retry logic with exponential backoff for transient failures
- Circuit breaker pattern for external service calls
- Database transaction management with rollback on error

Data integrity:

- Foreign key constraints enforce referential integrity
- Enrollment business rules validated in service layer
- Idempotency keys for critical operations (enrollment submission)
- Audit trail for all state changes

9.4 Maintainability

Code organization:

- Domain-driven design with clear boundaries
- Service layer abstraction separates business logic from controllers
- Shared validation schemas (Vine validators)
- Comprehensive OpenAPI documentation

Operational:

- Structured JSON logging with correlation IDs
- Feature flags for gradual rollout
- Blue-green deployment support
- Automated schema migration in CI/CD

9.5 Compliance

Data privacy (RA 10173):

- Purpose limitation for personal data collection
- Access controls restrict PII to authorized roles
- Data retention policies configurable
- Student consent management for optional uses

Academic records protection:

- FERPA-equivalent access restrictions
- Audit logging of grade access and modifications
- Secure transmission of transcripts to TORMS
- Public clearance verification without exposing PII

End of Architecture Design — USiS

Part 3 — Features & Requirements

USiS — University Student Information System Features & Requirements

Field	Value
Document ID	FEAT-USIS-001
Product	USiS
Companions	USiS-SRS.md · USiS-LLD.md
Date	November 2026

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1. Feature Summary

USiS provides comprehensive student information management across the complete academic lifecycle:

Area	Features	Status
Authentication	University Platform Access SSO integration, role-based routing	Delivered
Student Portal	Profile, enrollment workflow, schedule, assessments, COR, grades, history	Delivered
Faculty Portal	Class rosters, grade encoding/submission, advising, reports	Delivered
Registrar Portal	Applicant management, enrollment approval, academic configuration, reports	Delivered
Office Portals	Admissions, OSA, Guidance, Health Services, Scholarship management	Delivered
Assessment Integration	UFMS assessment/payment handoff with real-time status sync	Delivered
Grade Management	Encode, submit, corrections, GPA calculation, TORMS sync	Delivered
Clearance System	Multi-office clearance graph, certificate generation, QR verification	Delivered
Faculty Evaluation	Anonymous evaluation, aggregated results	Delivered
Reporting	Enrollment, grade, clearance, scholarship reports with export	Delivered

---

2. Feature Catalog

Feature ID	Feature	Description	Priority	Actors

F-AUTH-01	SSO Authentication	University Platform Access OIDC/JWT authentication	Must	All
F-AUTH-02	Role-based Routing	Automatic home path resolution (student/faculty/registrar/office)	Must	All
F-AUTH-03	Permission Enforcement	Granular permission checks on all API endpoints	Must	System
F-STU-01	Student Profile	View/edit demographic info, contact details, emergency contacts	Must	Student
F-STU-02	Enrollment Workflow	Search sections, add to cart, conflict detection, submit for approval	Must	Student
F-STU-03	Schedule Viewing	Calendar and list views of enrolled sections with details	Must	Student
F-STU-04	Assessment View	Line-item assessment breakdown from UFMS integration	Must	Student
F-STU-05	Payment Status	Real-time UFMS payment verification status display	Must	Student
F-STU-06	COR Generation	Download PDF Certificate of Registration after approval	Must	Student
F-STU-07	Grade Viewing	Posted grades by term with GPA calculation	Must	Student
F-STU-08	Academic History	Complete enrollment and grade history across all terms	Must	Student
F-STU-09	Support Requests	Submit help requests to offices (OSA, Guidance, Health, Scholarship)	Must	Student
F-STU-10	Organization Applications	Apply for student organization accreditation with attachments	Should	Student
F-STU-11	ID Card Request	Digital student ID request submission	Should	Student
F-STU-12	Queue Tickets	Obtain service queue numbers for office visits	Should	Student
F-STU-13	Faculty Evaluation	Anonymous faculty evaluation during evaluation windows	Must	Student
F-STU-14	Clearance Status	View office-by-office clearance breakdown	Must	Student
F-STU-15	Clearance Certificate	Download clearance certificate with QR verification code	Must	Student
F-FAC-01	Class Rosters	View enrolled students in assigned sections	Must	Faculty

F-FAC-02	Grade Encoding	Enter preliminary and final grades with validation	Must	Faculty
F-FAC-03	Grade Submission	Submit completed grades with digital confirmation	Must	Faculty
F-FAC-04	Grade History	View submission history and status by section	Must	Faculty
F-FAC-05	Grade Corrections	Request corrections after submission with justification	Should	Faculty
F-FAC-06	Roster Export	Download class rosters as CSV and PDF	Must	Faculty
F-FAC-07	Attendance Tracking	Mark student attendance by session	Should	Faculty
F-FAC-08	Advisee Management	View assigned advisees with enrollment status	Must	Faculty/Adviser
F-FAC-09	Advising Approval	Approve or recommend modifications to advisee enrollments	Must	Faculty/Adviser
F-FAC-10	Advising Notes	Add recommendations and session notes	Must	Faculty/Adviser
F-FAC-11	Evaluation Results	View anonymized aggregated evaluation results	Should	Faculty
F-REG-01	Applicant Management	View/search applicants from admission pipeline	Must	Registrar
F-REG-02	Student Admission	Admit applicants with student number and Identity provisioning	Must	Registrar
F-REG-03	Enrollment Review	View all enrollments with advanced filtering	Must	Registrar
F-REG-04	Enrollment Approval	Approve/reject enrollments with reason codes	Must	Registrar
F-REG-05	Batch Approval	Process multiple enrollments with criteria selection	Should	Registrar
F-REG-06	Business Rule Override	Override system rules with audit logging	Must	Registrar
F-REG-07	Academic Calendar	Configure terms, enrollment windows, deadlines	Must	Registrar
F-REG-08	Program Management	Define programs, curricula, courses	Must	Registrar
F-REG-09	Section Management	Create sections with instructor, schedule, room, capacity	Must	Registrar

F-REG-10	Prerequisite Rules	Configure prerequisite and co-requisite chains	Must	Registrar
F-REG-11	Grading System Config	Define grading policies and scales by program	Must	Registrar
F-REG-12	Enrollment Reports	Summary by program, status, term with export	Must	Registrar
F-REG-13	Grade Reports	Distribution, dean's list, probation lists	Must	Registrar
F-REG-14	Report Export	CSV, Excel, PDF export for all reports	Must	Registrar
F-REG-15	Grade Monitoring	Track submission status across sections	Must	Registrar
F-REG-16	Evaluation Management	Configure evaluation periods and questionnaires	Must	Registrar
F-REG-17	Evaluation Reports	View aggregated results with filtering	Should	Registrar
F-REG-18	Waitlist Management	Manage waitlists and seat releases	Should	Registrar
F-ADM-01	Entrance Exam Scheduling	Schedule exam sessions with capacity management	Must	Admissions
F-ADM-02	Exam Results	View exam scores and track admission decisions	Must	Admissions
F-OSA-01	Discipline Management	Manage discipline cases with confidential records	Must	OSA Staff
F-OSA-02	Organization Accreditation	Process student organization applications	Must	OSA Staff
F-OSA-03	Event Coordination	Manage campus events and participation tracking	Should	OSA Staff
F-GDN-01	Counseling Scheduling	Schedule confidential counseling sessions	Must	Guidance
F-GDN-02	Counseling Records	Maintain restricted-access counseling notes	Must	Guidance
F-GDN-03	Guidance Clearance	Issue guidance clearances	Must	Guidance
F-HLT-01	Medical Records	Maintain student health records	Must	Health Services
F-HLT-02	Vaccination Tracking	Track immunization requirements	Must	Health Services

F-HLT-03	Health Clearance	Issue health clearances	Must	Health Services
F-SCH-01	Scholarship Awards	Manage scholarship types and awards	Must	Scholarship
F-SCH-02	Disbursement Tracking	Track scholarship disbursement schedules	Must	Scholarship
F-SCH-03	Academic Requirements	Monitor scholarship academic standing requirements	Must	Scholarship
F-INT-01	Assessment Preparation	Generate tuition assessments from enrollment data	Must	System
F-INT-02	UFMS Assessment Sync	Push assessments to UFMS via internal API	Must	System
F-INT-03	Payment Status Query	Real-time UFMS payment verification	Must	System
F-INT-04	TORMS Grade Sync	Expose grade projections for TORMS consumption	Must	System
F-INT-05	Clearance Aggregation	Multi-system clearance status consolidation	Must	System
F-CLR-01	Clearance Requirements	Configure clearance items by office and term	Must	System
F-CLR-02	Clearance Certificate**	Generate certificates with QR codes	Must	System
F-CLR-03	Public Verification	Unauthenticated clearance verification endpoint	Must	Public
F-EVL-01	Evaluation Configuration	Setup periods, questionnaires, response scales	Must	Registrar
F-EVL-02	Evaluation Submission	Students evaluate faculty anonymously	Must	Student
F-EVL-03	Result Anonymization	Ensure no student-response linkage in results	Must	System
F-EVL-04	Statistical Aggregation	Compute evaluation statistics by section/faculty	Must	System
F-RPT-01	Enrollment Summary	Reports by term, program, level	Must	Registrar
F-RPT-02	Grade Distribution	Reports by course and section	Must	Registrar
F-RPT-03	Dean's List	Academic honor roll generation	Must	Registrar
F-RPT-04	Graduation Candidates	Eligible graduates by term	Must	Registrar

F-RPT-05	Report Export Audit	Log all report exports for compliance	Must	System

---

3. Requirements Traceability Matrix

3.1 Functional Requirements Mapping

Requirement ID	Requirement	Related Features	Priority	Implementation Status
FR-AUTH-001	SSO via University Platform Access JWKS	F-AUTH-01	Must	Delivered
FR-AUTH-002	Role-based home path resolution	F-AUTH-02	Must	Delivered
FR-AUTH-003	Permission enforcement on all endpoints	F-AUTH-03	Must	Delivered
FR-AUTH-004	MFA support for privileged roles	F-AUTH-01	Must	Identity-delegated
FR-AUTH-005	Graceful Identity service degradation	F-AUTH-01	Must	Delivered
FR-STU-001..019	Student portal features	F-STU-01..15	Must/Should	Delivered
FR-FAC-001..012	Faculty portal features	F-FAC-01..11	Must/Should	Delivered
FR-REG-001..018	Registrar portal features	F-REG-01..18	Must/Should	Delivered
FR-OFF-001..015	Office portal features	F-ADM-01..02, F-OSA-01..03, F-GDN-01..03, F-HLT-01..03, F-SCH-01..03	Must/Should	Delivered
FR-PAY-001..007	Assessment/payment integration	F-INT-01..03	Must/Should	Delivered
FR-GRD-001..007	Grade management and sync	F-FAC-02..05, F-INT-04	Must	Delivered
FR-CLR-001..005	Clearance management	F-CLR-01..03, F-INT-05	Must	Delivered
FR-EVL-001..006	Faculty evaluation	F-EVL-01..04, F-STU-13, F-FAC-11	Must/Should	Delivered
FR-RPT-001..006	Reporting and analytics	F-RPT-01..05	Must	Delivered

3.2 Non-Functional Requirements Mapping

NFR Category	Requirement ID	Related Features	Implementation Approach
Performance	NFR-PERF-001..006	All features	Database indexing, caching, query optimization, background jobs
Scalability	NFR-SCAL-001..004	All features	Horizontal API scaling, read replicas, stateless design
Security	NFR-SEC-001..010	F-AUTH-*, F-INT-*	TLS, RBAC, rate limiting, encryption, audit logging
Reliability	NFR-REL-001..005	All features	HA deployment, backups, DR procedures, graceful degradation
Maintainability	NFR-MAIN-001..005	All features	Modular architecture, OpenAPI docs, structured logging, versioned migrations
Compliance	NFR-COMP-001..005	Student data features	RA 10173 compliance, purpose limitation, consent management, FERPA-equivalent

3.3 Source Requirements Traceability

WMSU TOR VI.E.1 compliance matrix:

TOR Requirement	Feature IDs	Implementation Notes	Compliance Status
Applicant management and admissions	F-REG-01, F-REG-02, F-ADM-01, F-ADM-02	Portal handles entrance exam delivery; USiS receives results	Delivered
Student registration and enrollment	F-STU-02, F-REG-03, F-REG-04	Complete workflow with advising, payment, approval gates	Delivered
Subject/section scheduling	F-REG-09, F-STU-03, F-FAC-01	Real-time conflict detection, capacity enforcement	Delivered
Grade encoding and management	F-FAC-02, F-FAC-03, F-REG-15	Faculty encode, registrar monitors, TORMS sync	Delivered
Student academic records	F-STU-07, F-STU-08, F-INT-04	USiS is SoR; TORMS consumes projections	Delivered
Assessment and payment coordination	F-INT-01, F-INT-02, F-INT-03	UFMS integration; no local payment invention	Delivered
Clearance processing	F-CLR-01, F-CLR-02, F-CLR-03, F-STU-14, F-STU-15	Multi-office graph, QR verification	Delivered
Faculty evaluation	F-EVL-01, F-EVL-02, F-STU-13	Anonymous submissions, aggregated results	Delivered

Student support services	F-STU-09, F-OSA-01..03, F-GDN-01..03, F-HLT-01..03, F-SCH-01..03	Multi-office portal with permission packs	Delivered
Institutional reporting	F-RPT-01..05, F-REG-12..14	CSV/Excel/PDF export with audit logging	Delivered

Deferred/Out of scope:

- Public guest application UI → Unified Access Portal responsibility
- Financial transaction processing → UFMS responsibility
- TOR issuance and alumni registry → TORMS responsibility
- Library, documents, inventory → ULMIS, UDMS, UIMS

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4. Non-Functional Requirements Summary

4.1 Performance Targets

Metric	Target	Measurement
API response (p95)	≤2s standard, ≤5s enrollment	APM monitoring
Concurrent users	≥500 active sessions	Load testing
Health check	≤500ms	Uptime monitoring
Report generation	≤10s standard	Query optimization

4.2 Security Controls

Control	Implementation	Verification
TLS enforcement	Reverse proxy/ingress	SSL Labs scan
RBAC	Permission checks on all endpoints	Automated security tests
Rate limiting	Redis-backed rate limiter	Load test verification
Audit logging	Structured logs for privileged ops	Log review

4.3 Reliability Measures

Measure	Target	Method
Availability	≥99.5% during enrollment	SLA monitoring
Backup frequency	Daily incremental, weekly full	Automated backup verification
RPO	≤15 minutes	Tested recovery procedures

Graceful degradation	UFMS/TORMS outage handling	Circuit breaker testing

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5. Actors & Feature Access

Actor	Primary Features	Restricted Features
Student	F-STU-01..15, F-EVL-02	Own data only; cannot view other students
Faculty	F-FAC-01..11	Assigned classes only
Adviser	F-FAC-08..10	Assigned advisees only
Registrar	F-REG-01..18, F-RPT-01..05	System-wide academic data
Admissions Officer	F-ADM-01, F-ADM-02	Applicant records
OSA Staff	F-OSA-01..03	Student conduct records
Guidance Counselor	F-GDN-01..03	Confidential counseling data
Health Services	F-HLT-01..03	Student health records
Scholarship Officer	F-SCH-01..03	Scholarship data
Public	F-CLR-03	Clearance verification only
System (Background)	F-INT-01..05, F-RPT-05	Automated integrations

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End of Features & Requirements — USiS

Part 4 — Low-Level Design (LLD) Workflows

USiS — University Student Information System Low-Level Design

Field	Value
Document ID	LLD-USIS-001
Product	USiS
Companion	USiS-SRS.md
Date	November 2026

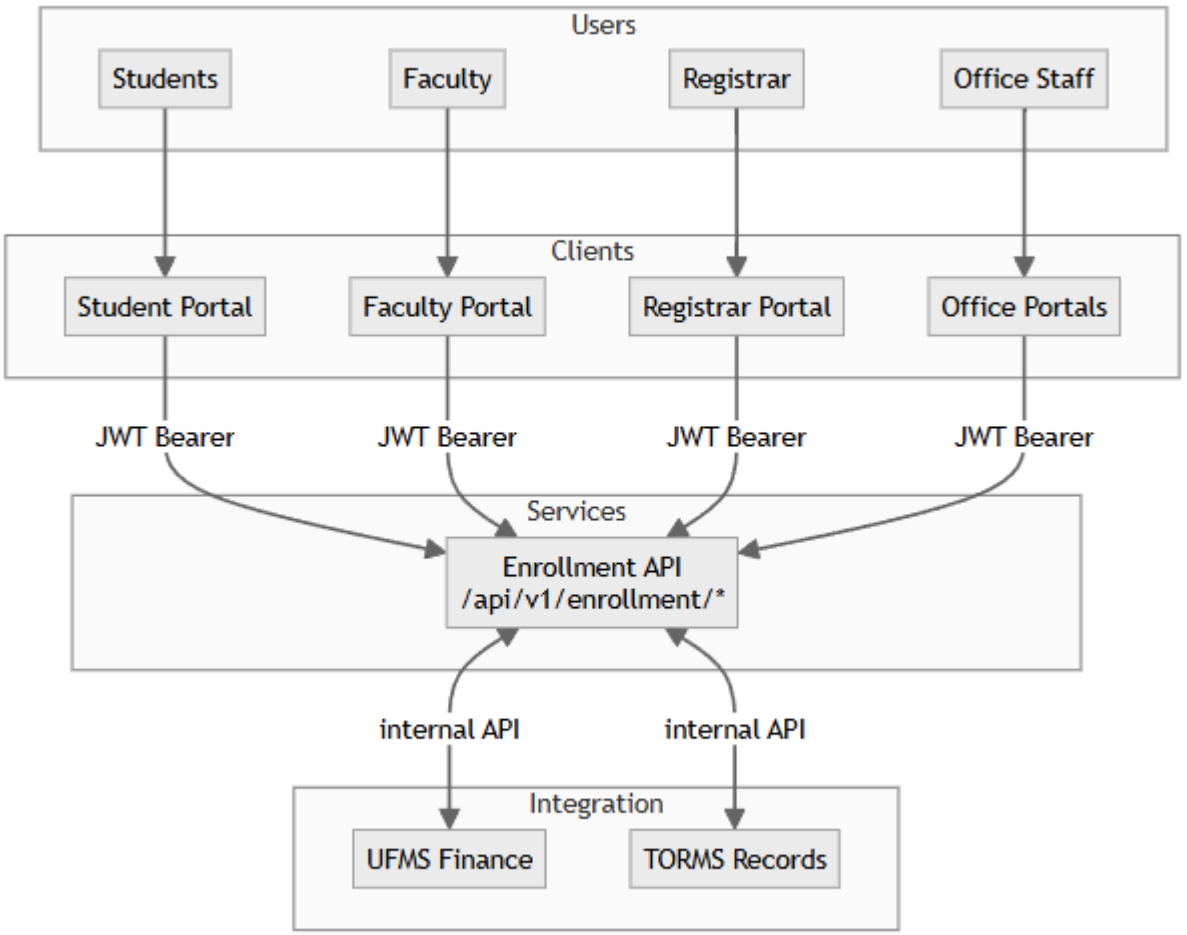
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1. Purpose

This Low-Level Design documents critical workflows for USiS enrollment, grade management, advising, assessment/payment integration, clearance processing, and faculty evaluation — providing sequence diagrams, state machines, and implementation details for development and QA teams.

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2. System Context (Multi-Portal Architecture)



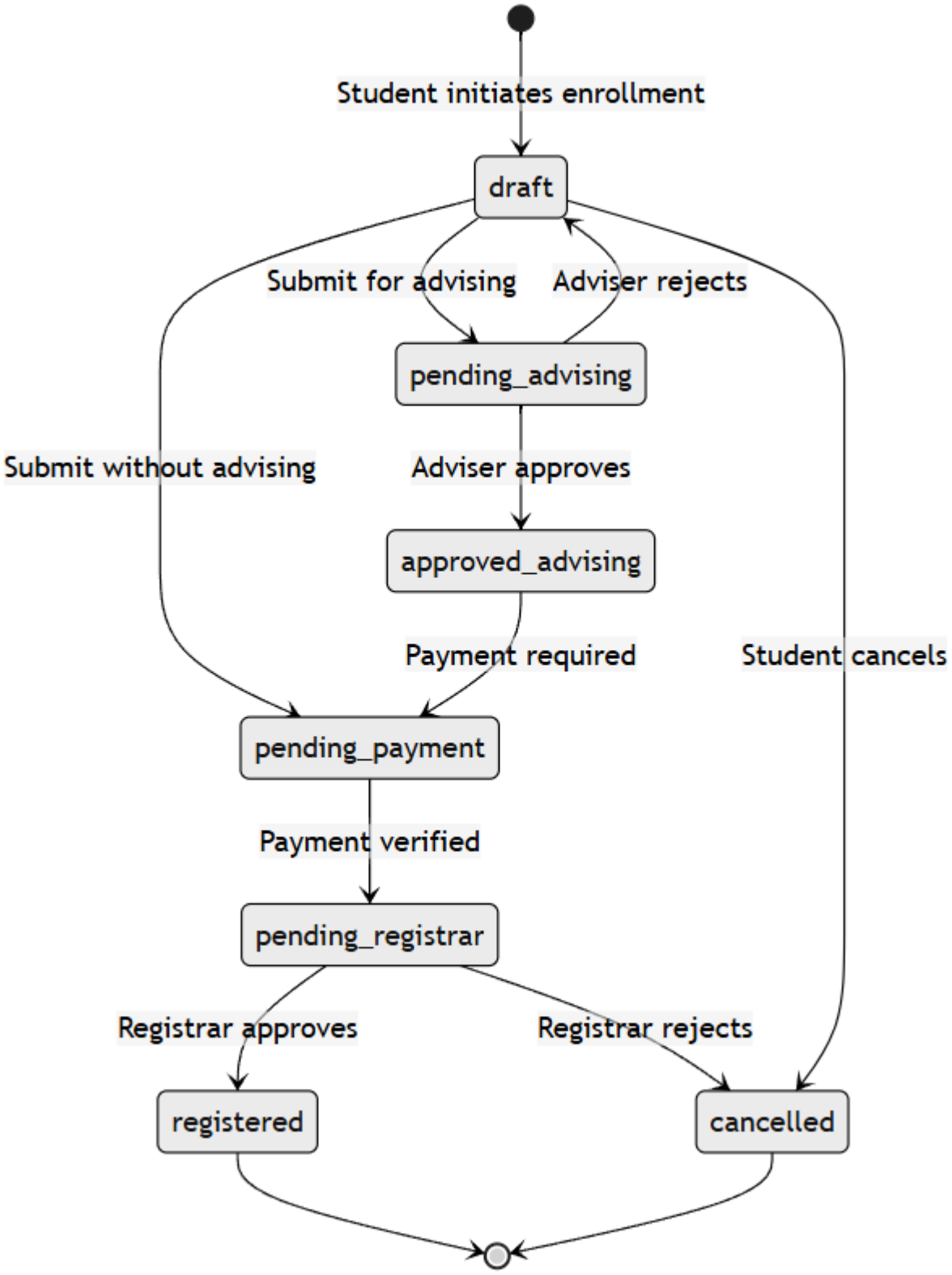
Client-specific routes:

- Student: `/student/*`
- Faculty: `/faculty/*`
- Registrar: `/registrar/*`
- Offices: `/offices/:officeType/*`

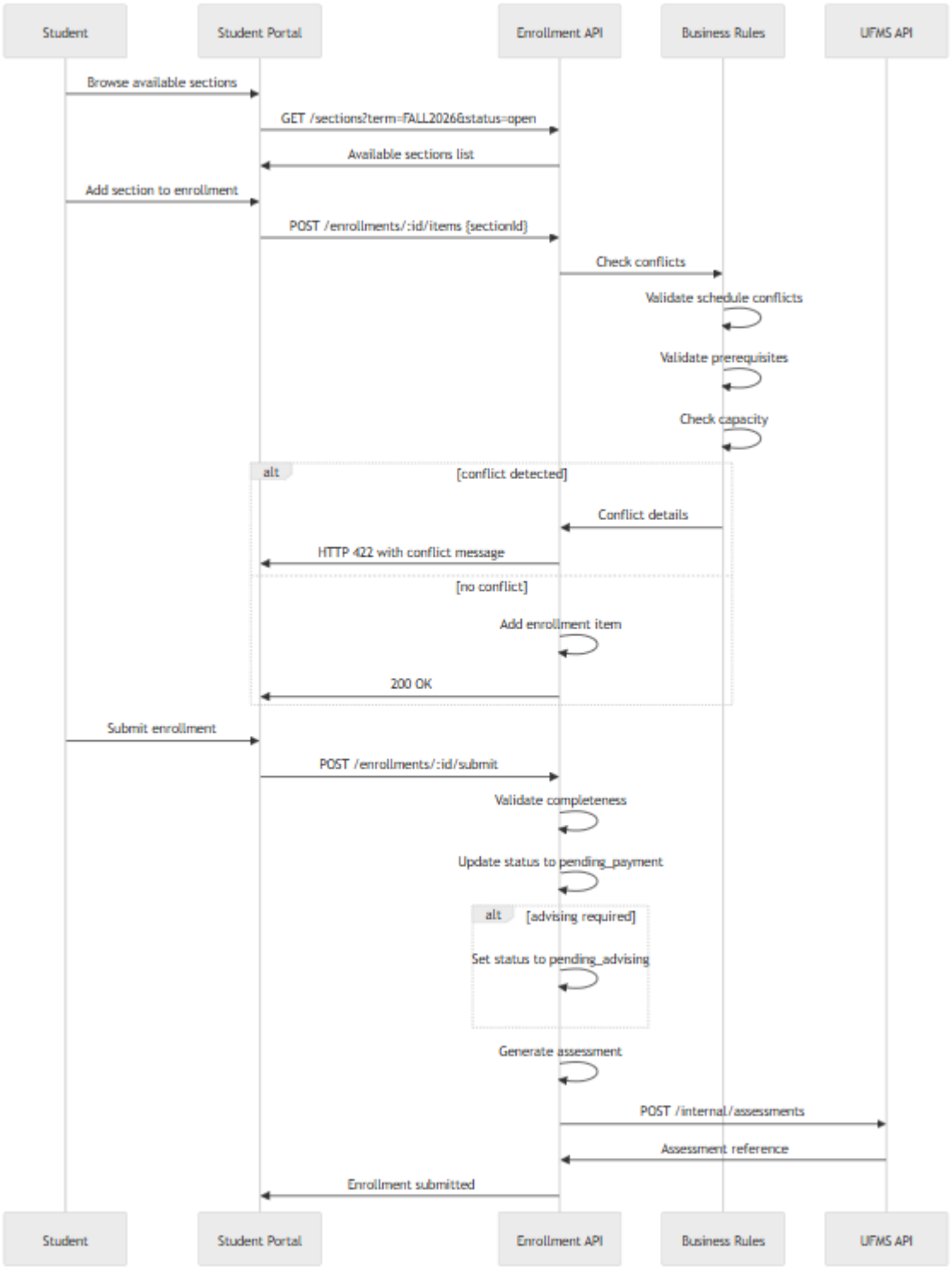
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3. Student Enrollment Workflow

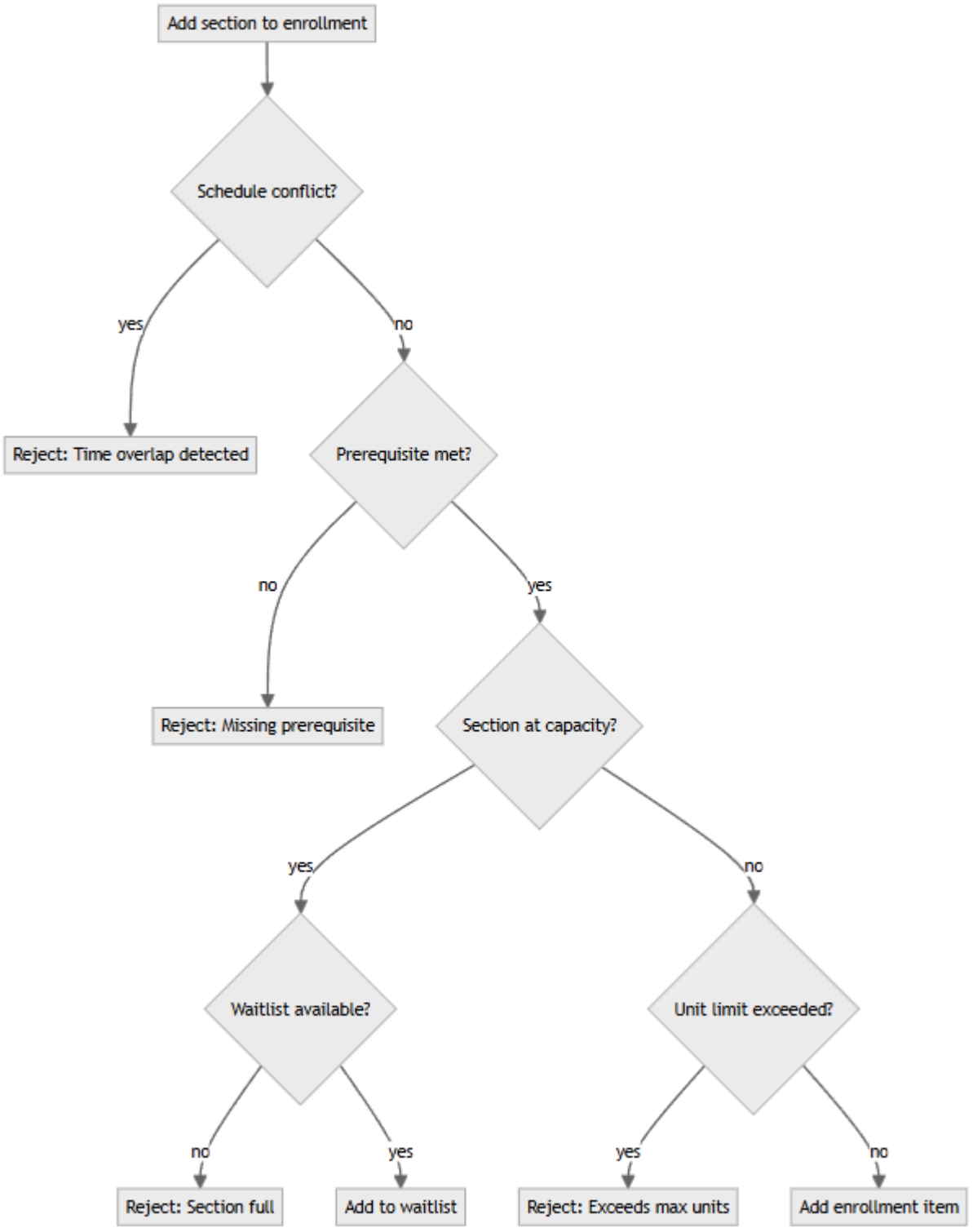
3.1 Enrollment State Machine



3.2 Enrollment Submission Sequence



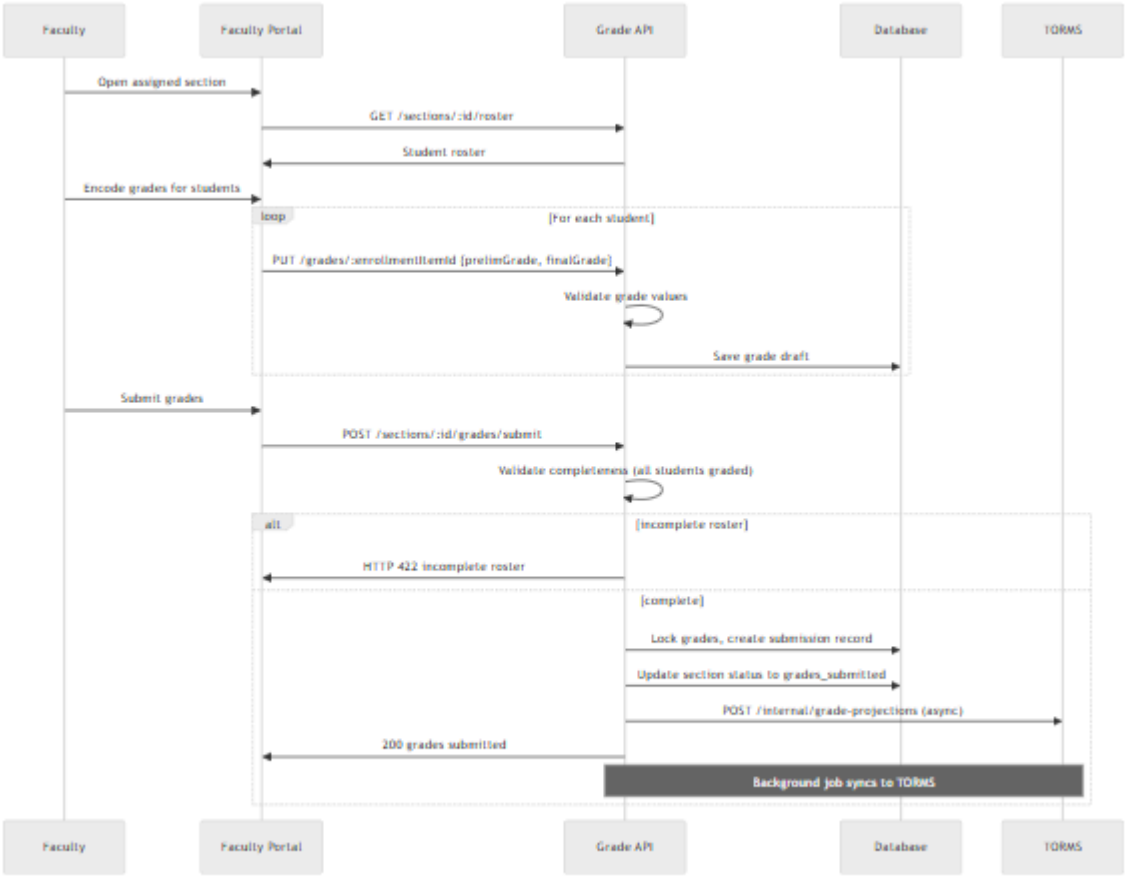
3.3 Conflict Detection Logic



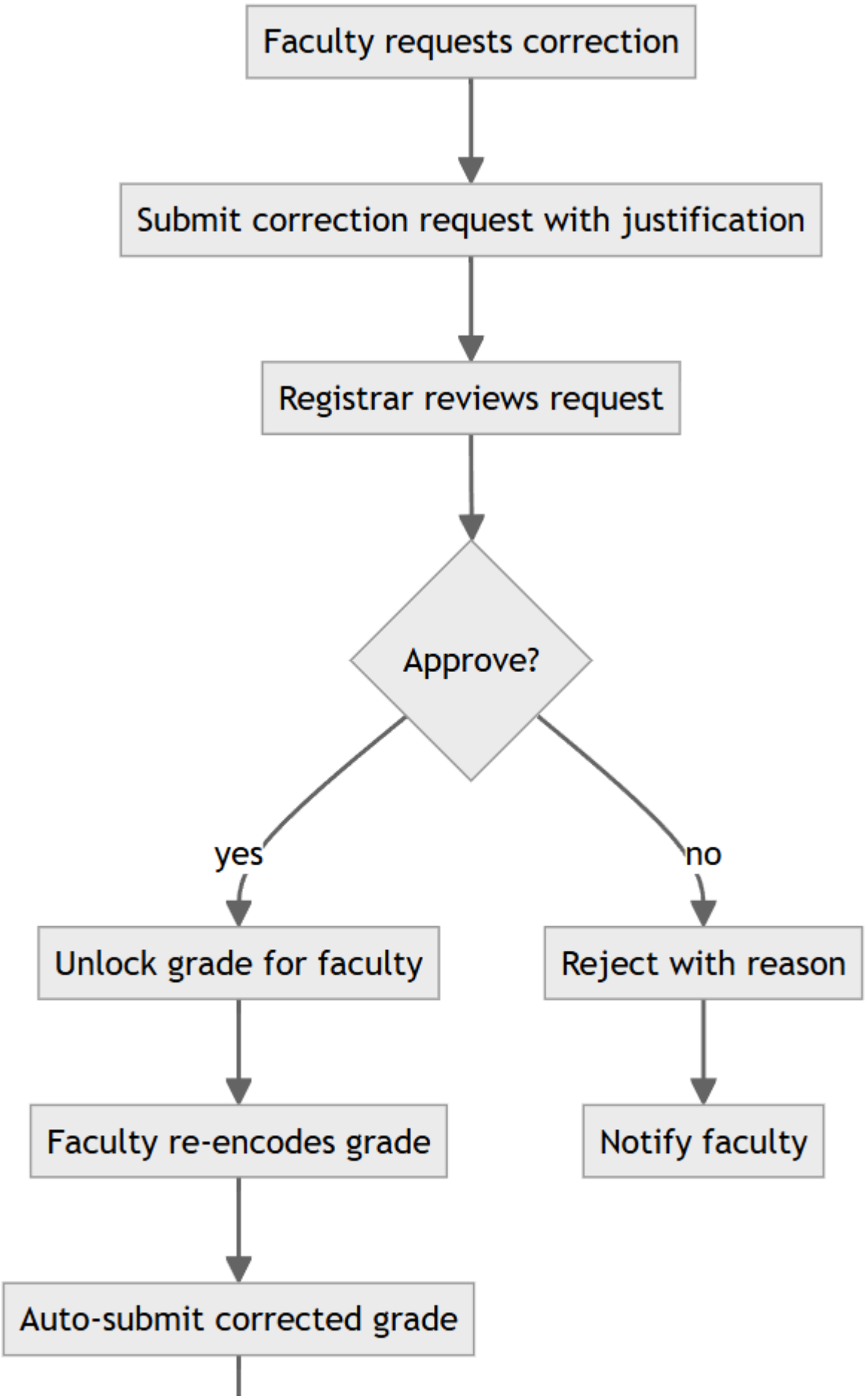
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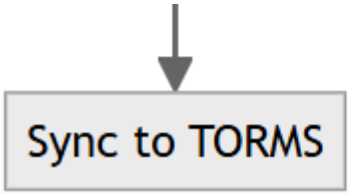
4. Grade Management Workflow

4.1 Faculty Grade Submission

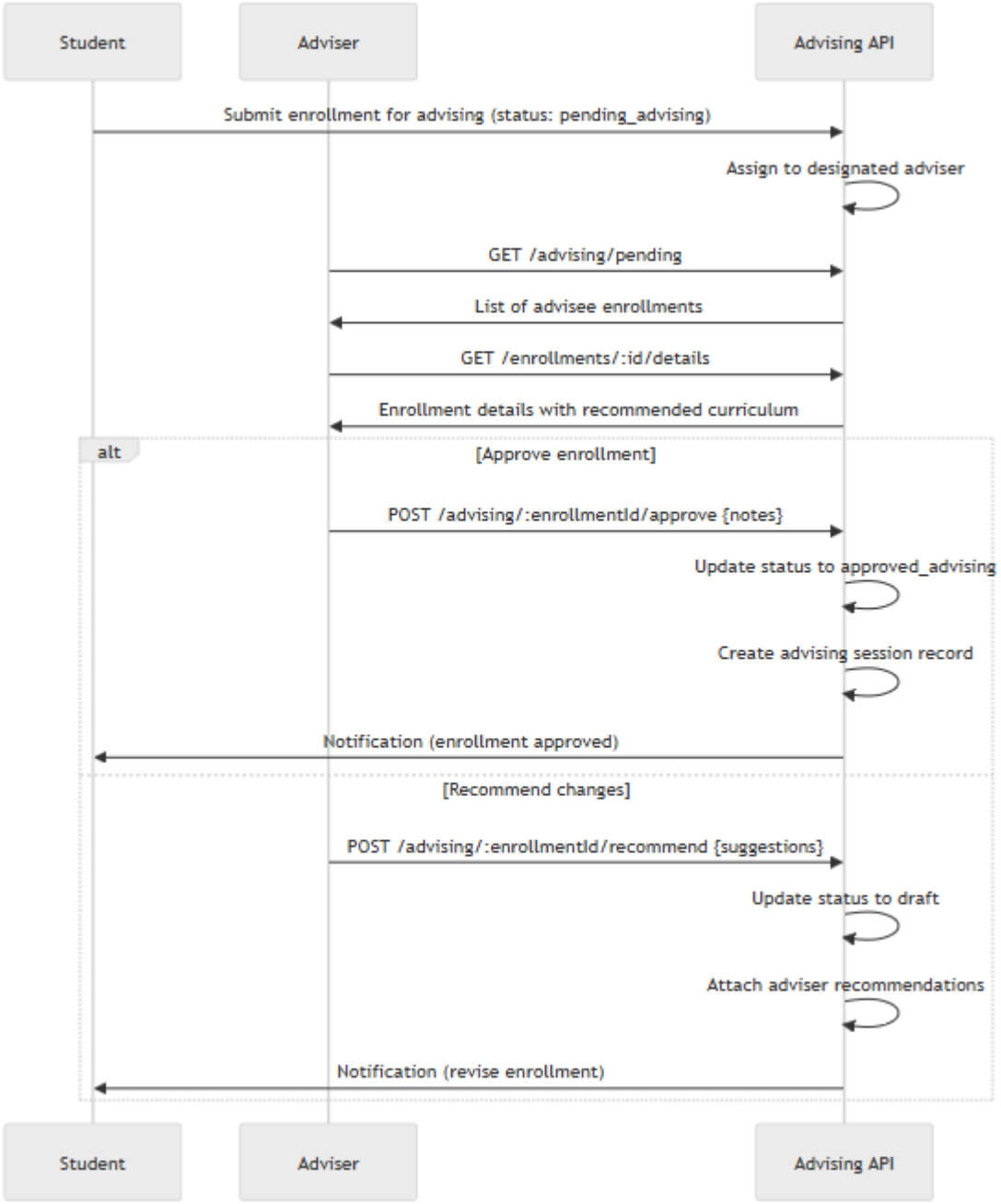


4.2 Grade Correction Workflow


Faculty requests correction
Submit correction request with justification
Registrar reviews request
Approve?
yes
no
Unlock grade for faculty
Reject with reason
Faculty re-encodes grade
Notify faculty
Auto-submit corrected grade

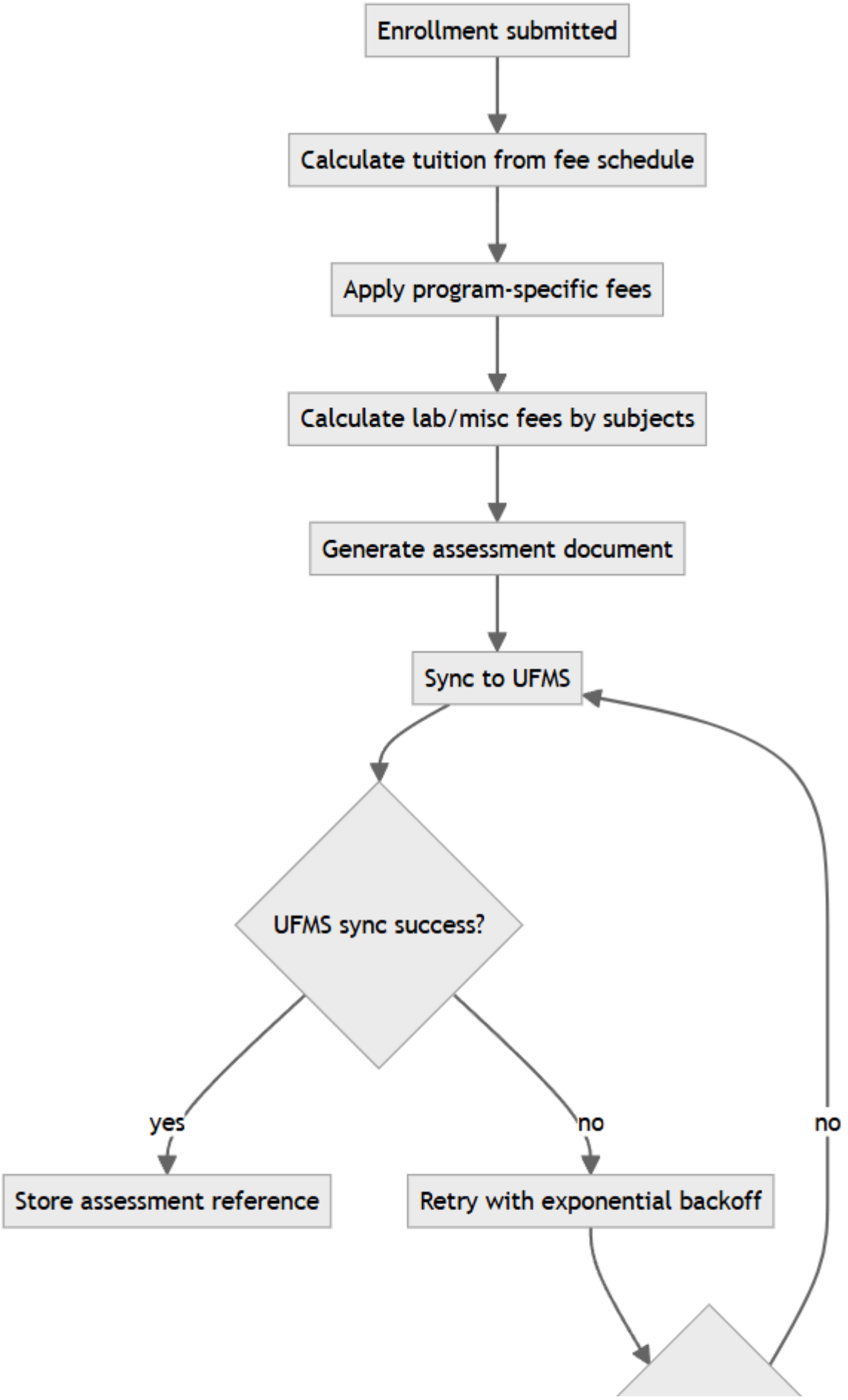


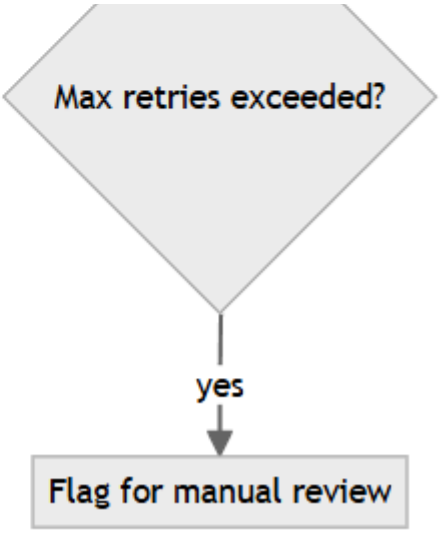
5. Advising Workflow



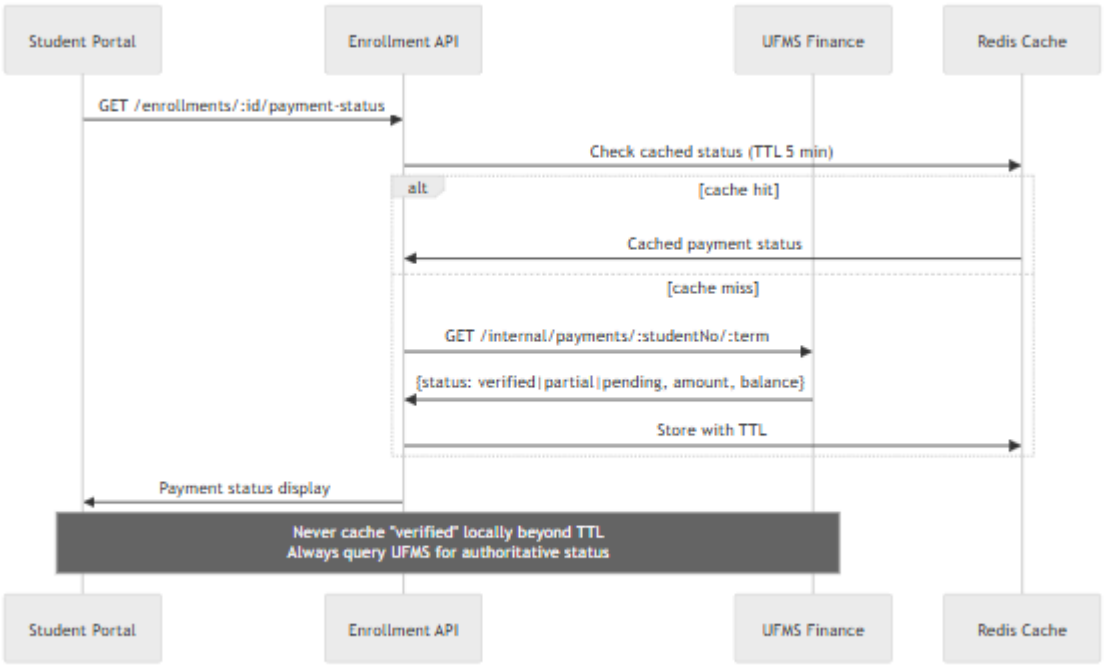
6. Assessment & Payment Integration

6.1 Assessment Preparation


Enrollment submitted
Calculate tuition from fee schedule
Apply program-specific fees
Calculate lab/misc fees by subjects
Generate assessment document
Sync to UFMS
UFMS sync success?
yes
no
Store assessment reference
Retry with exponential backoff
no



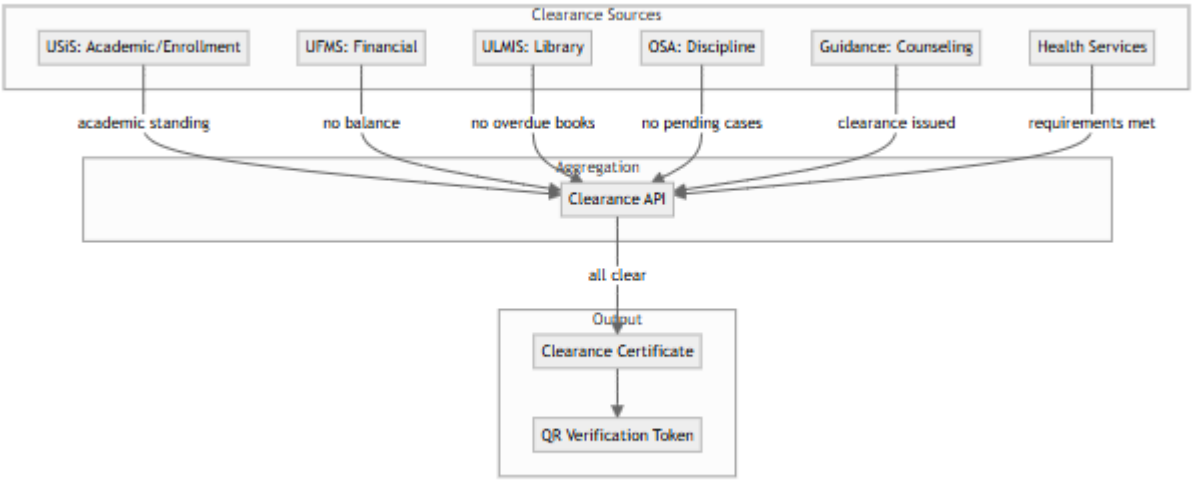
6.2 Payment Status Verification



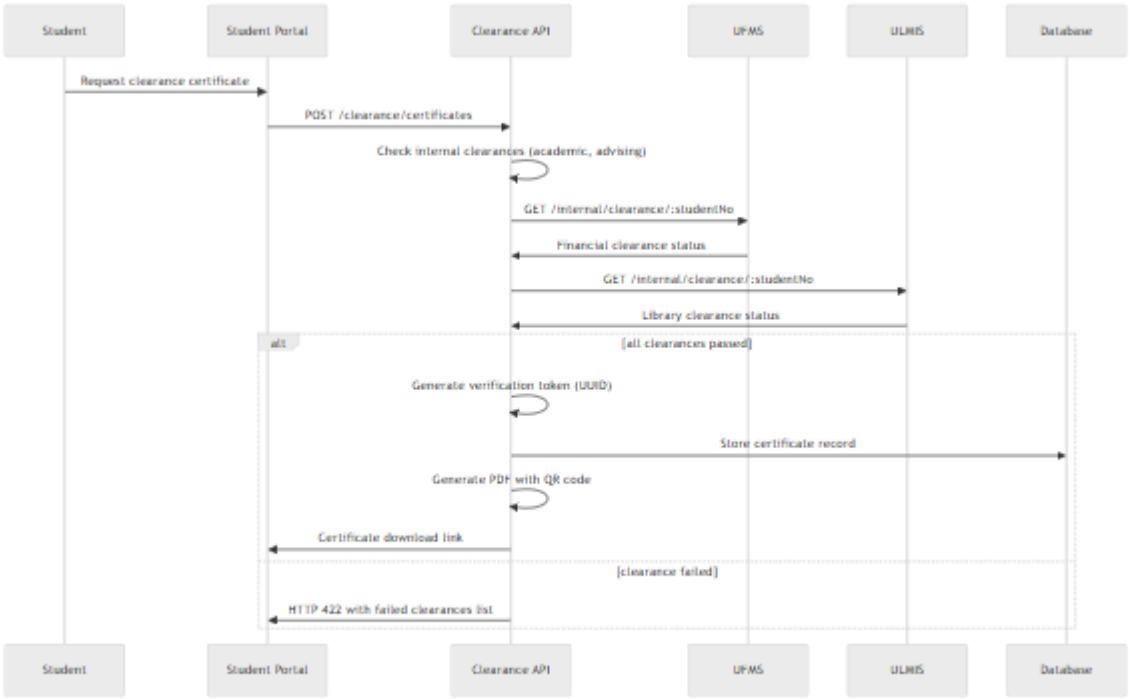
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7. Clearance Management

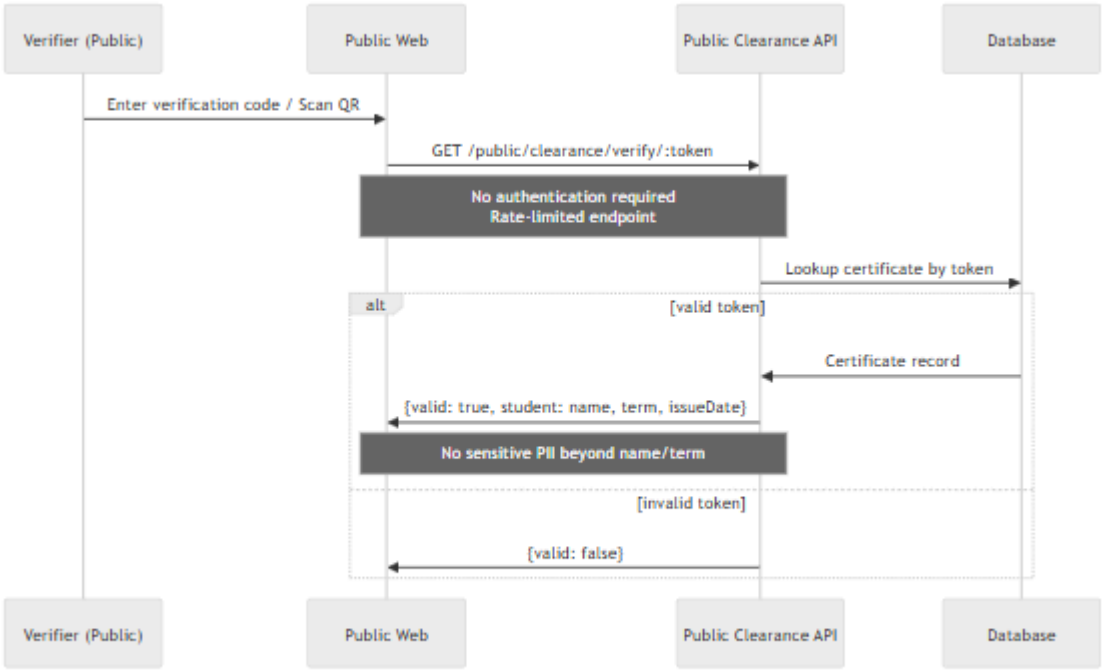
7.1 Clearance Status Aggregation



7.2 Clearance Certificate Generation



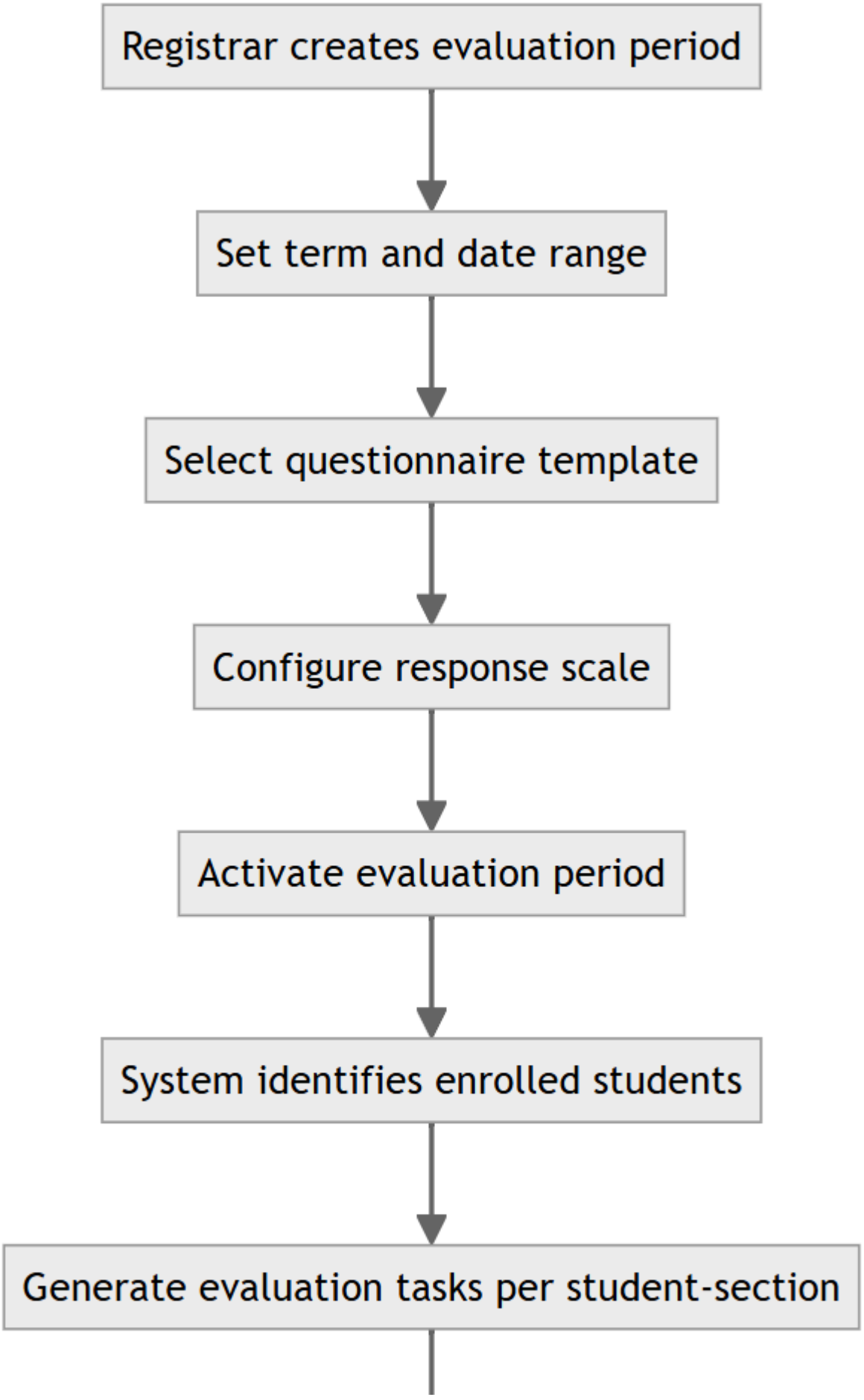
7.3 Public Clearance Verification

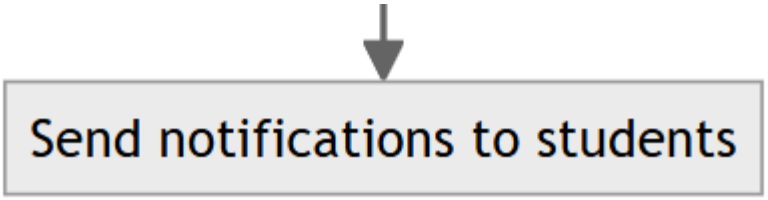


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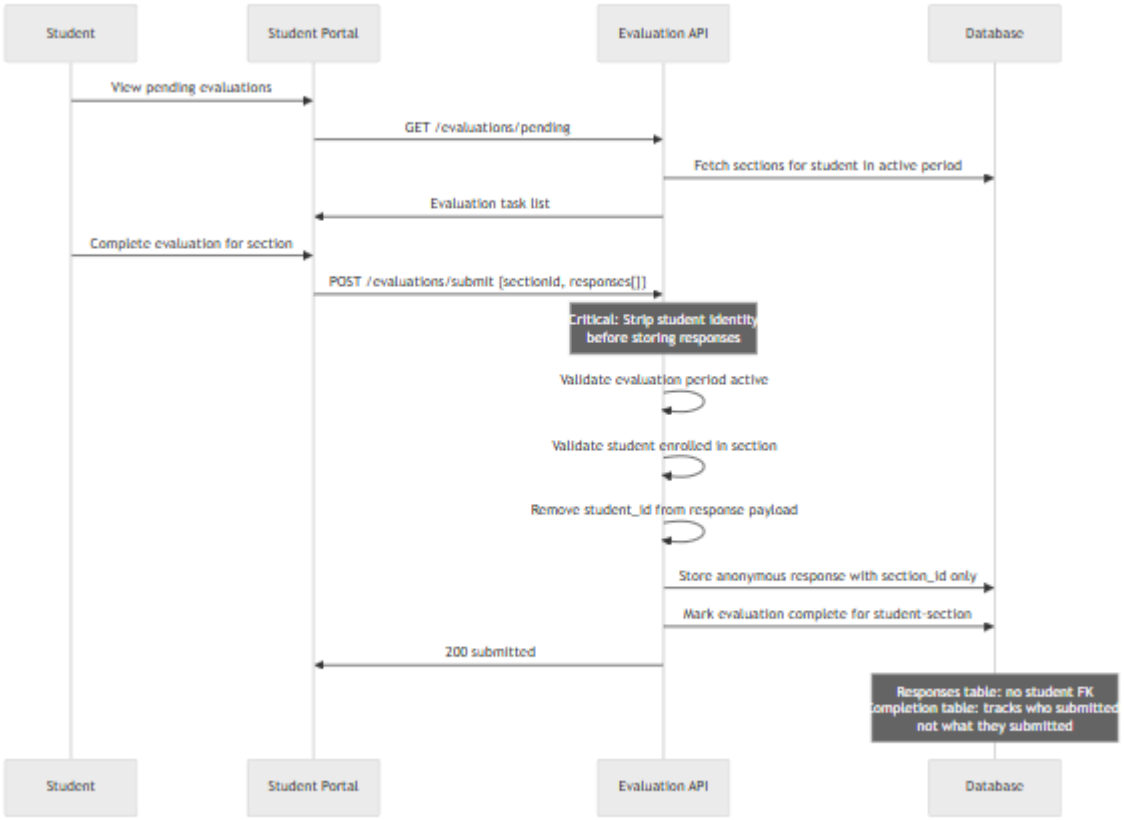
8. Faculty Evaluation Workflow

8.1 Evaluation Period Setup

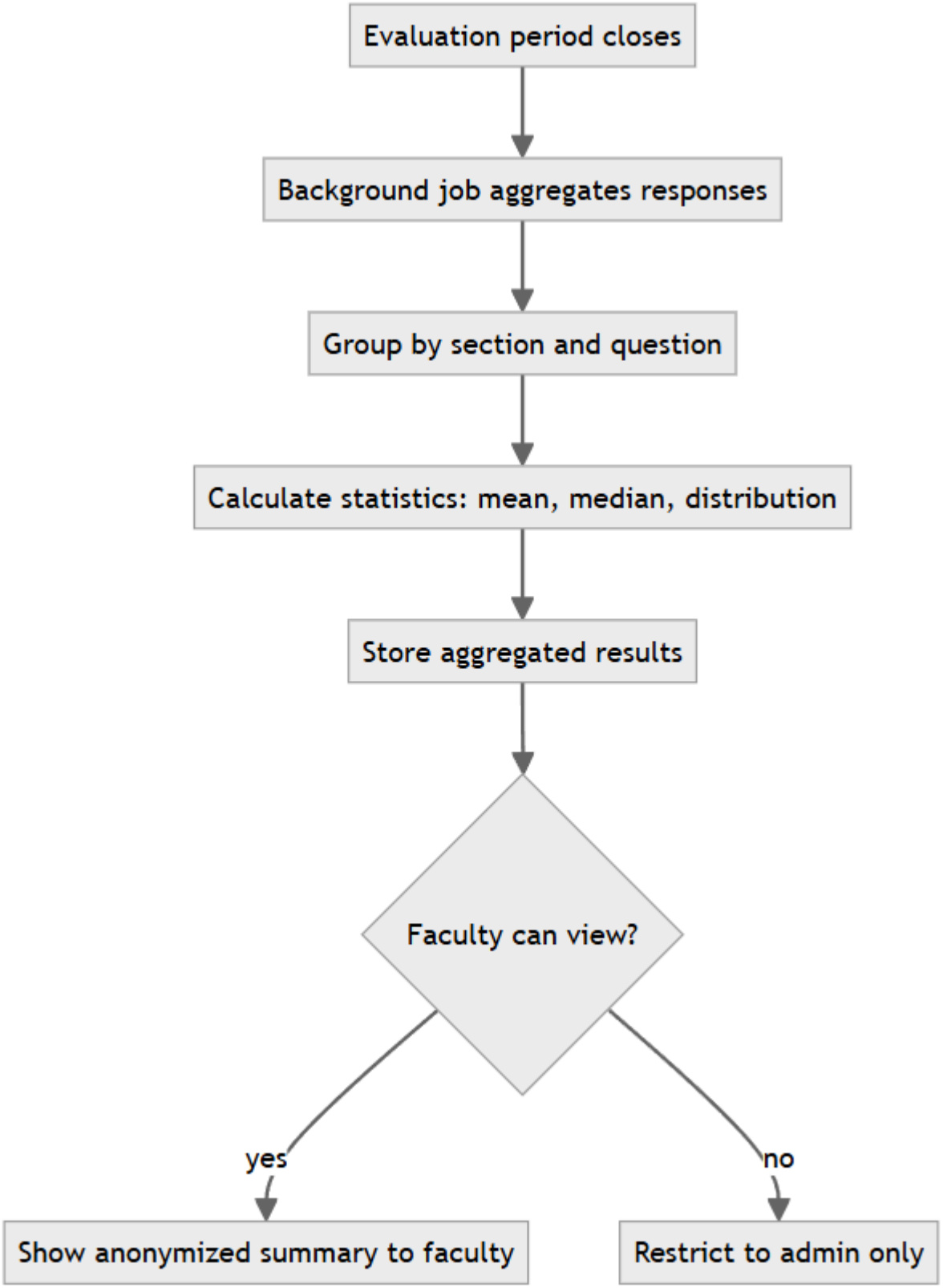

Registrar creates evaluation period
Set term and date range
Select questionnaire template
Configure response scale
Activate evaluation period
System identifies enrolled students
Generate evaluation tasks per student-section



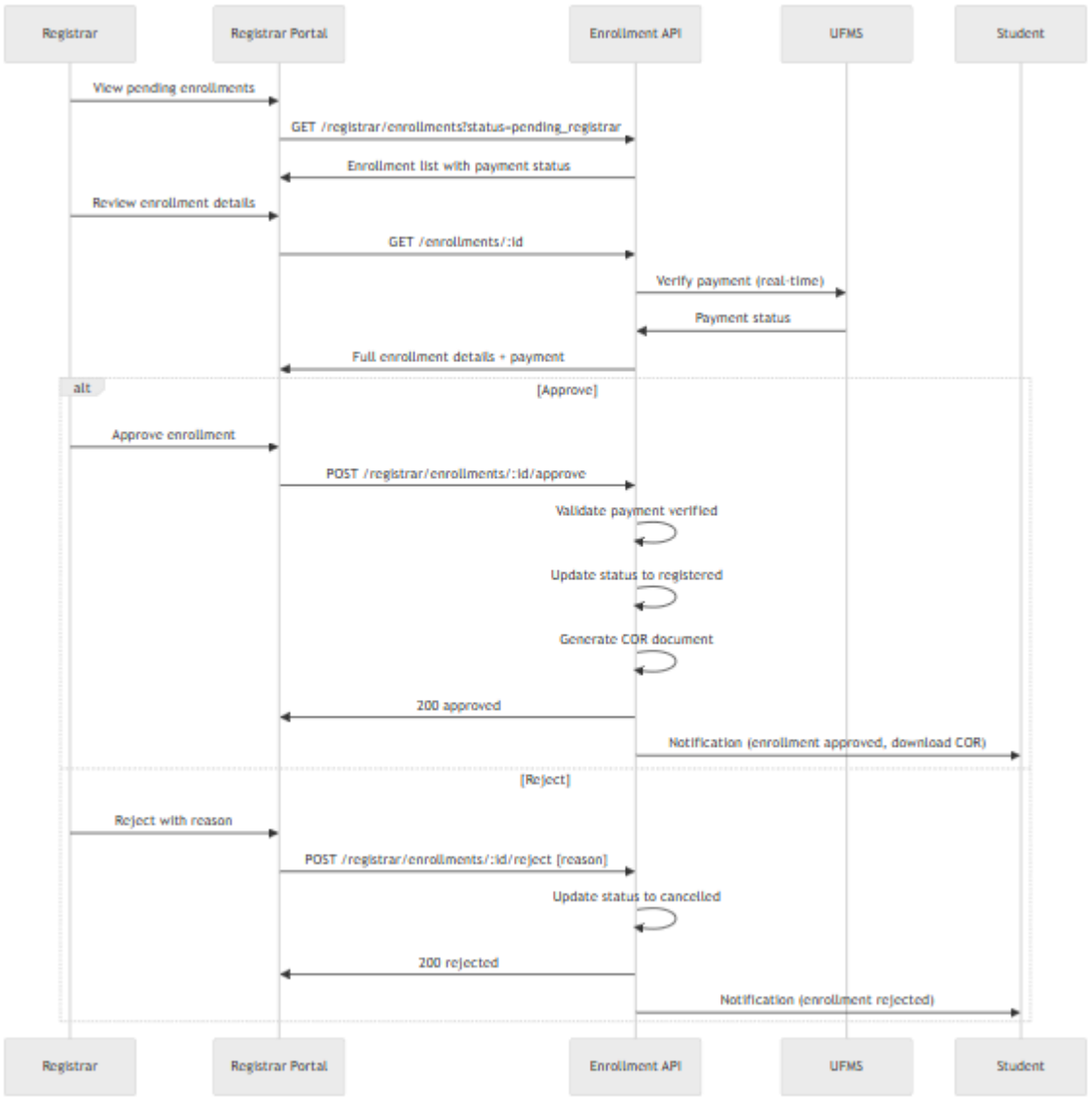
8.2 Anonymous Submission



8.3 Result Aggregation

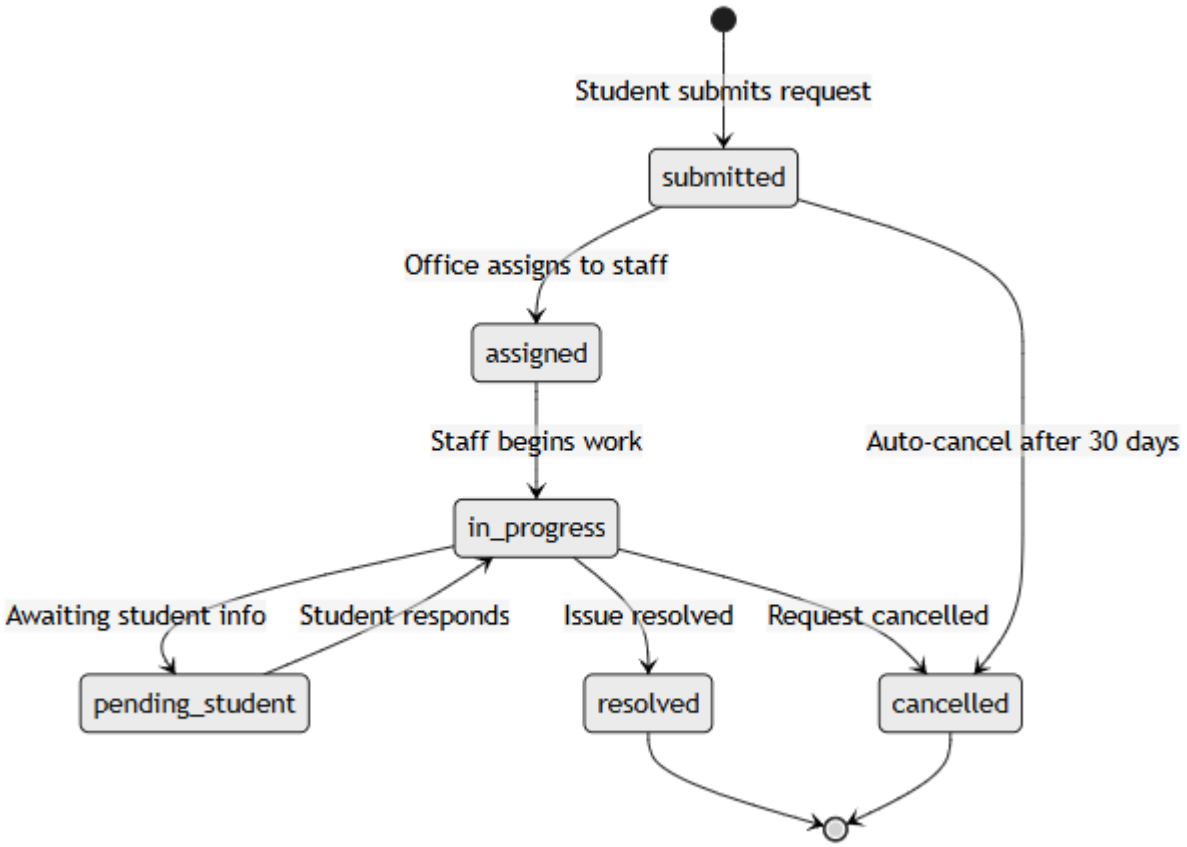

Evaluation period closes
Background job aggregates responses
Group by section and question
Calculate statistics: mean, median, distribution
Store aggregated results
Faculty can view?
yes
no
Show anonymized summary to faculty
Restrict to admin only

9. Registrar Enrollment Approval



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10. Office Support Request Workflow



11. Error Cases & Edge Conditions

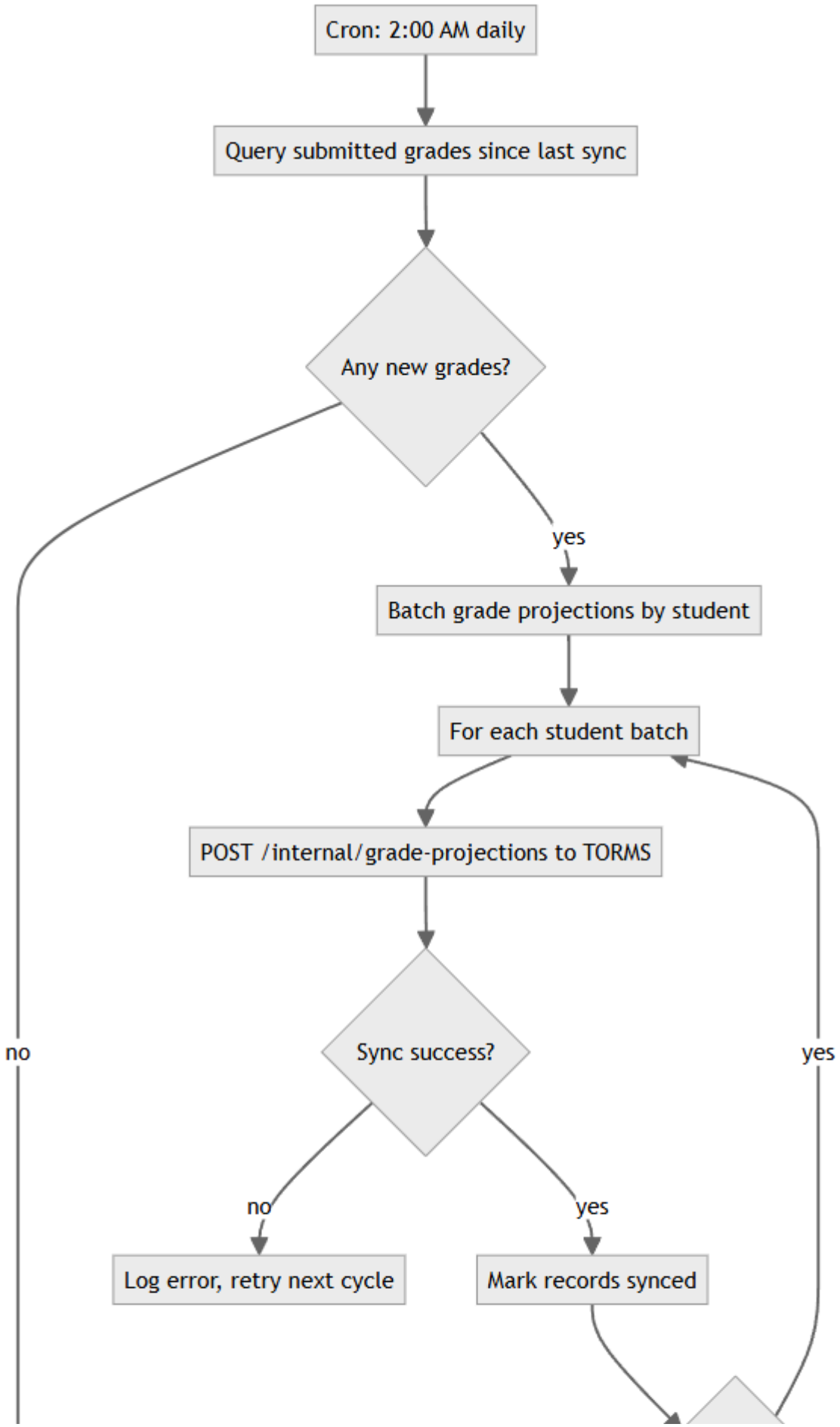
Case	Expected Behavior	HTTP Status	User Message
Enroll with schedule conflict	Block enrollment, return conflicting section	422	"Schedule conflict with [Section Name] on [Days] at [Time]"
Enroll with missing prerequisite	Block enrollment, list missing prerequisites	422	"Prerequisite required: [Course Code]"
Enroll in full section without waitlist	Block enrollment	422	"Section is full. No waitlist available."
Submit incomplete grade roster	Block submission	422	"Cannot submit. [N] students have no grades."
Query UFMS when service down	Return cached status with warning	200	"Payment status (last checked [timestamp]). Verify with cashier if recently paid."

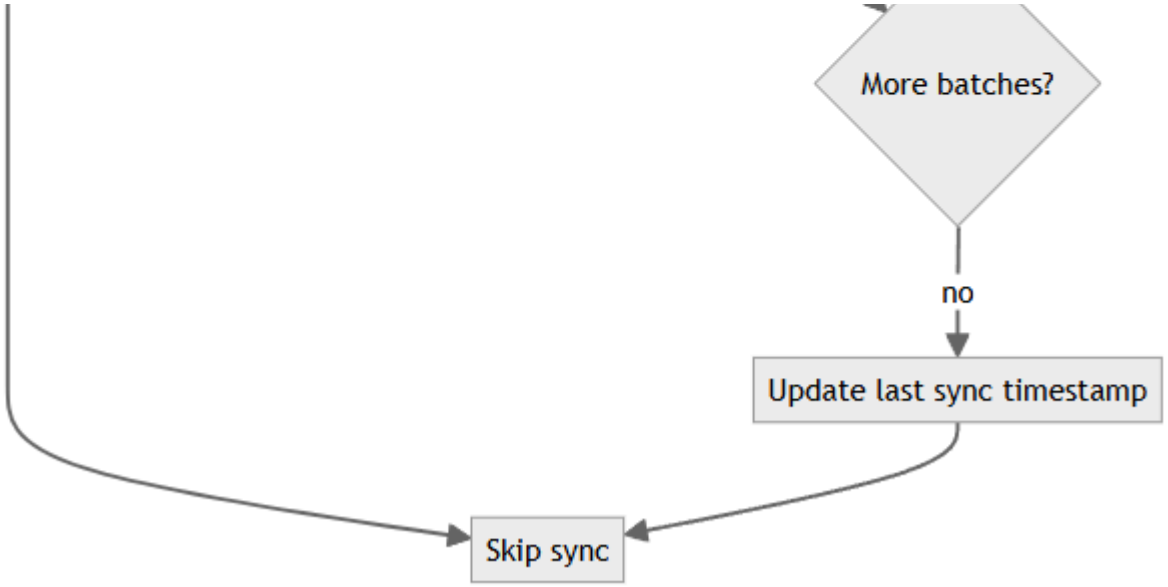
Generate clearance with ULMIS down	Fail certificate generation	503	"Clearance unavailable. Library system temporarily unreachable."
Verify invalid clearance token	Return not found	404	"Invalid verification code."
Faculty view grades before submission	Allow view of draft grades	200	Display with "Draft" indicator
Student access other student's data	Authorization failure	403	"Access denied."
Registrar override without permission	Authorization failure	403	"Insufficient permissions for override."
Evaluation submission outside period	Block submission	422	"Evaluation period has ended."
Duplicate enrollment submission (idempotency)	Return original submission result	200	Previous submission details

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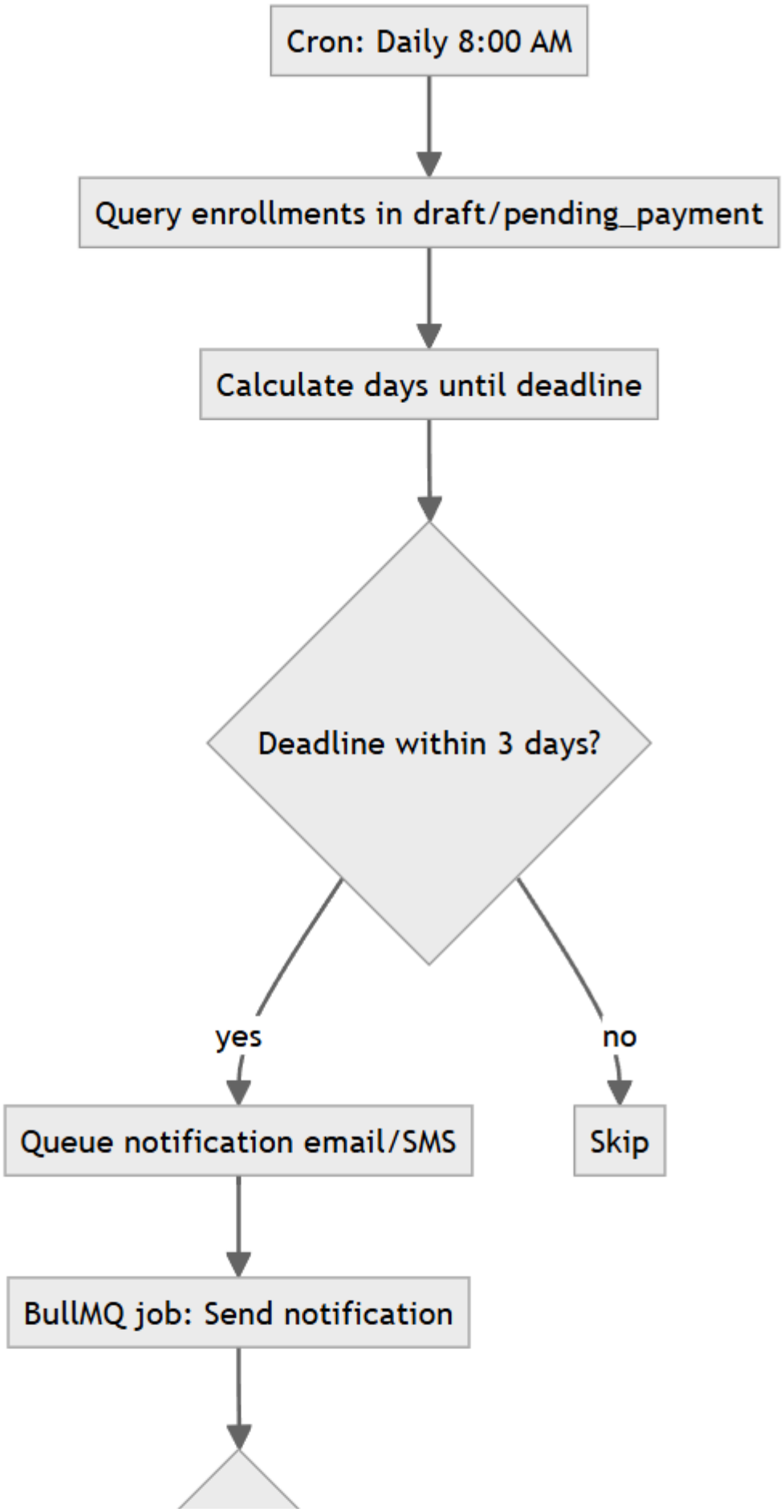
12. Background Job Workflows

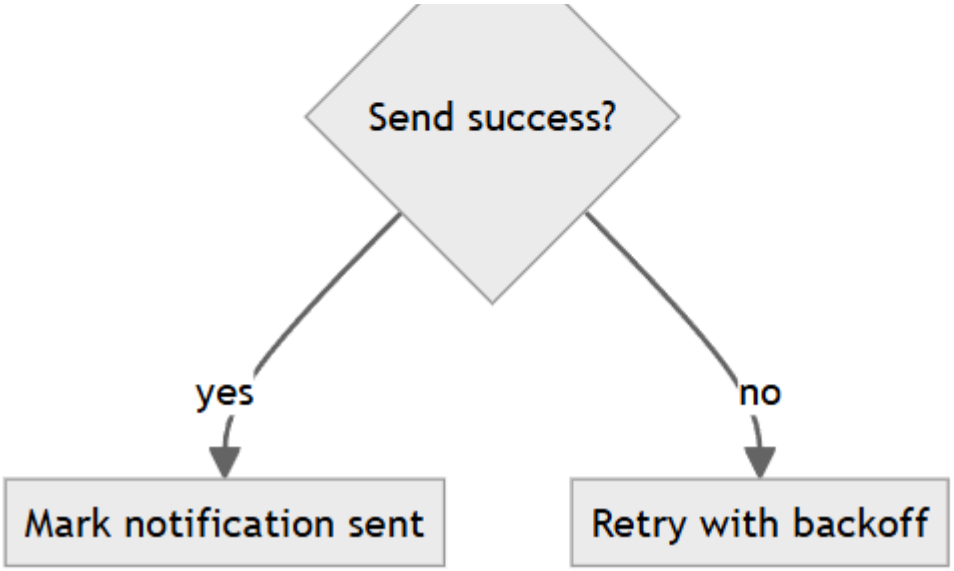
12.1 Nightly Grade Sync to TORMS


Cron: 2:00 AM daily
Query submitted grades since last sync
Any new grades?
yes
Batch grade projections by student
For each student batch
POST /internal/grade-projections to TORMS
Sync success?
no
yes
no
yes
Log error, retry next cycle
Mark records synced



12.2 Enrollment Deadline Notifications


Cron: Daily 8:00 AM
Query enrollments in draft/pending_payment
Calculate days until deadline
Deadline within 3 days?
yes
no
Queue notification email/SMS
Skip
BullMQ job: Send notification



End of LLD — USiS
